

Social Preferences: Fundamental Characteristics and Economic Consequences[†]

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We review the vast literature on social preferences by assessing what is known about their fundamental properties, their distribution in the broader population, and their consequences for important economic and political behaviors. We provide, in particular, an overview of the empirical characteristics of distributional preferences and how they are affected by merit, luck, and concerns for equality of opportunity. In addition, we discuss the evidence for reciprocity and guilt aversion and assess the empirical relevance of self-image and social image concerns in prosocial behaviors. The overall evidence indicates that a large majority of individuals have some sort of social preferences, while purely self-interested subjects are a minority. We also document converging insights from the lab and the field on the impact of wage inequality on work morale, employees' resistance to wage cuts, and the role of social preferences for cooperation and collective action, distributive politics, and individuals' selection into different occupations. (JEL D63, D71, D72, D86, H23, J53)

1. Introduction

The topic of social preferences has received a great deal of attention in economics and in the broader social science literature in the past three decades. A large and still-growing literature has documented the existence of social preferences, their social and economic implications, and the various conditions under which they influence the equilibria and outcomes of human interactions. An individual, i , has social preferences if the utility function of i , U_i , is not merely a function of i 's material payoff, π_i , in which case it could be written as $U_i(\pi_i)$, but also depends on the material payoff of at least one other player, j , π_j , such that

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$U_i(\pi_i, \pi_j)$. Theories of social preferences differ in how exactly π_j affects player i 's utility—for example, through social comparisons between π_i and π_j or through the perceived kindness associated with particular strategies that generate π_i and π_j —however, they all involve the assumption that other players' material payoffs enter i 's utility function in some way. This means that individuals with social preferences are willing to sacrifice money or other material resources to increase other players' payoffs (if the marginal utility of π_j is positive) or to hurt them (if π_j yields negative marginal utility), which has potentially far-reaching consequences for many practical applications. In this paper, we will summarize what has been learned in this line of research and we will identify what needs to be known and done to make further progress.

The role of social preferences can, in principle, manifest itself in myriads of human interactions. Behaviors associated with social preferences range from helping friends and neighbors to support for coworkers, from charitable donations to participation in collective actions against oppressive dictatorships, from individuals' support for sustainable investments to workers' responses to wage inequality in companies, from affluent individuals' support for taxing the rich to workers' collective stance against wage cuts. However, a multitude of factors other than social preferences also play a role in all these cases. For example, expectation of future material benefits or reputational concerns can influence individuals' behavior in these situations. Therefore, a considerable part of the research on social preferences is based on incentivized laboratory experiments where one can rule out these concerns with certainty by ensuring that the experiments are one-shot and that the parties interact anonymously with each other.¹ More recently, however, we also experienced a healthy transition to more field evidence—evidence we will also cover in this review.

Due to the vast literature that has accumulated over the last three decades, it is useful to provide some guidance about the structure of this review (see Box 1). We will describe the evidence for the existence of social preferences gathered in canonical (“paradigmatic”) experiments in Section 2. However, readers who are already familiar with these findings may directly move on to one of the other sections of the paper, as we have tried to keep the individual sections relatively self-contained. For example, readers who are mainly interested in the economic and political consequences of social preferences may proceed to Section 6, which discusses rich evidence from the field and the lab on these issues.

The paradigmatic experiments discussed in Section 2, while indicating the existence of social preferences and capturing essential components of important real-life interactions, are a poor tool for the precise identification of the properties of social preferences. Moreover, these experiments led to a fundamental puzzle because the same individuals sometimes displayed very selfish behavior, while they behaved very prosocially at other times. Section 3 discusses the theories of distributional preferences that helped resolve this puzzle by providing a unifying explanation of these seemingly disparate facts, which then also led to a rich empirical literature on the distribution of distributional preferences in the broader population.

Section 3 takes the recent literature on this topic into account and discusses what we know about the key motivational drivers underlying distributional preferences and the relative

¹The implementation of incentivized anonymous one-shot experiments is thus a frequently employed conservative strategy that makes sense, but it is also important to keep in mind that important aspects of social relations and their influence on social preferences may be lost in these experiments (Frohlich, Oppenheimer, and Kurki 2004). It is, however, possible to introduce many real-life features (e.g., lack of anonymity, reputation formation, the role of merit and entitlements, etc.) into the lab and to study them in a controlled manner.

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frequencies of the different distributional preference types. This section also documents one of the key findings of the recent literature that a considerable majority of individuals in the broader population has other-regarding distributional preferences in the form of inequality aversion and altruism while predominantly selfish individuals constitute a minority. In addition, Section 3 explores how notions of entitlement, merit, and luck affect distributional preferences and how psychologists and economists examined and modeled these factors. This section also indicates that considerable progress has been made, including interesting insights into the large cultural differences in how luck affects individuals' fairness views and how cognitive biases shape individuals' perception of merit.

Section 4 discusses evidence that theories of distributional preferences can hardly explain; this evidence led to the development of social preferences theories such as reciprocity theory and guilt aversion theory where the other player's material payoff enters *i*'s utility function through notions of kindness and guilt. A key characteristic of these approaches is that kindness and guilt are defined in terms of individuals' first- and second-order beliefs about (others') behaviors and the implied payoffs. This poses challenges in identifying these preferences because the beliefs are endogenous. This section will also provide a critical discussion on the extent to which these challenges have been met and on the empirical relevance of the notions of reciprocity and guilt aversion in the lab and the field.

People may not only care directly about others' payoff or others' deservingness and kindness, but they may also care about their own identity and how others perceive them—which leads us to theories of self-image and social image. We examine these theories, the extent to which the evidence supports them, and some of their economic implications in Section 5 of the paper.

Section 6 documents the widespread and far-reaching economic and political consequences of social preferences. In this context, we discuss their important role in human cooperation; their effects on labor relations including the potential consequences on macroeconomic phenomena; their role in affecting incentives, contractual arrangements, and institutions; as well as their effects on politics. Finally, we discuss important open questions—such as the determinants of social preferences or the relationship between social norms and social preferences—that offer exciting research opportunities in Section 7.

2. *Social Preferences and Self-Interest in Paradigmatic Economic Experiments*

There are several paradigmatic experiments that, when played one-shot and under anonymity between the players involved, document the widespread existence of social preferences. We call these experiments “paradigmatic” because they all capture an essential feature of an important economic or social situation. The experiments are described in more detail in Box 2 on “Paradigmatic Economic Experiments.” All of these experiments have been replicated dozens of times (or more) and show robust, replicable behavioral patterns (see Camerer 2003 for a review). The critical player in these experiments (i.e., the person for whom one can measure social preferences) always has a simple and transparent money maximizing choice, while systematic (i.e., nonrandom) deviations from that choice indicate some form of social preference.

For example, many responders in the ultimatum experiment from societies across the globe indicate a social preference by rejecting uneven offers and earning nothing as a consequence (Güth, Schmittberger, and Schwarze 1982; Henrich et al. 2001). The proposer often anticipates this responder behavior, inducing him or her to make relatively fair offers so that the responder appropriates on average roughly 40 percent of the pie even though the self-interest model predicts that he or she should get 0 percent.

In the dictator experiment, many subjects in the role of dictators make positive unilateral transfers—on average typically around 20 percent of the available money—to the recipient (Forsythe et al. 1994; Camerer 2003). Subjects in the role of workers in the gift exchange experiment (Fehr, Kirchsteiger, and Riedl 1993) respond to higher wages with higher costly effort levels, although the minimal effort would always be the money-maximizing choice. In the trust experiment (Berg, Dickhaut, and McCabe 1995), the trustees respond to trustors' positive transfers with positive back-transfers, although the money-maximizing choice would be to transfer back nothing. In social dilemma experiments (Dawes, McTavish, and Shaklee 1977; Dawes 1980) and public good experiments with (complete) defection as a dominant strategy (Andreoni 1988), many subjects nevertheless cooperate and make positive contributions to the public good.

And third parties in third-party punishment experiments (Fehr and Fischbacher 2004) who observe that other individuals have been treated unfairly punish the perpetrators at a cost to themselves. Interestingly, many of these other-regarding behaviors have also been observed at very high-stakes levels of up to \$10,000 (Slonim and Roth 1998; Cameron 1999; Fehr, Tougareva, and Fischbacher 2014; Larney, Rotella, and Barclay 2019; Dwyer et al. 2023).

Box 2. Paradigmatic Economic Experiments

In a *dictator experiment* one player—the dictator—is given a sum of money that she can unilaterally allocate between herself and a passive recipient who cannot make a decision. Self-interest predicts zero transfers to the recipient. In an *ultimatum experiment*, a proposer can make a single proposal of how to distribute a given sum of money between herself and a responder. The responder can accept or reject the proposed allocation. If she rejects, both players receive nothing. If she accepts, the proposed allocation is implemented. Self-interest predicts that the proposer makes the lowest possible money offer, which the responder will then accept. In a *third party punishment experiment*, two players, the dictator A and the recipient B, participate in a dictator game. A third player, the potential punisher C, observes how much A gives to B; then C can spend a proportion of his endowment on punishing A. A third party with a punishment option can, in principle, be added to any constituent base game such as, for example, the prisoners' dilemma game. Selfish third parties will never punish.

In a *trust experiment*, two players, A and B, each have an identical initial endowment. First, A decides whether to keep his endowment or to send some or all of it to B. Then B observes A's action and decides whether to keep the amount she receives or share some of it with A. The experimenter increases A's transfer by some proportion, so that both players are better off collectively if A transfers money and B sends back a sufficient amount. If both players are self-interested, B will always send back zero money, which will be anticipated by A, and therefore A will also send no money. In a *gift exchange experiment*, a subject in the role of an employer can offer a fixed wage to a subject in the role of a worker. After observing the wage, the worker chooses a costly effort level that raises the overall surplus that can be distributed among the parties. Self-interest predicts the lowest possible wage offer and the lowest possible effort level. In a generic *linear public goods experiment*, players have a token endowment they can invest in any proportion in a private project in their own favor or a public project where the proceeds are shared equally among the group members. Investment into the public project maximizes the group's aggregate earnings because the experimenter increases any donation to the public project, but each individual has a dominant money-maximizing strategy to invest the whole endowment into the private rather than the public project. All games described in this paragraph are examples of *social dilemma experiments*, where non-selfish cooperation causes a higher aggregate payoff for the group, but selfish behavior prevents this from happening.

In the *market experiment with responder competition*, the proposer decides how to split a given sum of money between herself and one of the competing responders. All $n > 1$ responders have to decide simultaneously whether to accept or reject the proposal. If all reject, all parties receive zero payoff; if some responders accept, one of them is randomly chosen to receive the proposed amount and all other responders receive zero. Self-interest predicts that the proposer makes the lowest possible money offer, which all responders will then accept. In the *market experiment with proposer competition*, all proposers simultaneously make a proposal of how to split a given sum of money with a single responder who can accept one of the proposals or reject all of them. The accepted proposal is implemented, while all other proposers earn zero. If all proposals are rejected, all players earn zero payoff. Self-interest predicts that competition among proposers drives the money offer up to the highest possible level.

The predictions for the experimental games mentioned above assume that the games are played one-shot and that interactions are anonymous.

The robust documentation of other-regarding behaviors in experiments has led many observers to study the extent and the way in which people care for other people's payoffs (Thibaut and Kelley 1959; Messick and McClintock 1968; Loewenstein, Thompson, and Bazerman 1989). However, one fundamental question that has remained unanswered for a long time is how one can reconcile the existence of social preferences with evidence from other paradigmatic experiments that appear to indicate that people are largely selfish. How can we explain this without arbitrarily assuming that individuals' preferences are different across experiments? For example, in a market experiment with responder competition (see Box 2), there is not just one responder as in the bilateral ultimatum experiment but several, so that the responders compete with each other for the share of the surplus the proposer offers. In this experiment, the parties behave in a much more self-interested manner, compared to the bilateral ultimatum experiment: the proposers make much more unequal offers, and the rejection of these uneven offers decreases dramatically, allowing the proposer to appropriate the lion's share of the surplus (Fischbacher, Fong, and Fehr 2009).

In a market experiment with proposer competition, one adds several competing proposers to a bilateral ultimatum experiment. Here again, the players behave much more in line with the self-interest prediction because the competition among the proposers drives up their offers such that the responder appropriates almost the whole surplus, that is, responders accept a distribution of payoffs (Roth et al. 1991; Fischbacher, Fong, and Fehr 2009) that appears very unfair. These facts are also consistent with the observations from competitive double auctions and competitive posted offer markets that indicate that the observed prices and quantities traded in these markets tend to quickly converge to the competitive equilibrium derived from selfish preferences (Smith 1982). Similarly, cooperation rates are often extremely low in the final period of finitely repeated public goods experiments, seemingly indicating that social preferences may play no role when sufficient learning has been possible.

How is the claim that many people have social preferences compatible with these facts? Does a little bit of learning or competition wipe out social preferences, that is, do they simply vanish under competitive pressure? Even worse, is behavior in these paradigmatic experiments that document social preferences just a strange behavioral aberration or can models of social preferences also explain the conditions under which other-regarding individuals behave as if they were completely self-interested? Section 3 will discuss how these puzzles could be solved by distributional theories of social preferences.

3. *Fundamental Properties of Distributional Preferences*

3.1 *Social Preferences over Payoff Distributions*

3.1.1 *Models of Distributional Preferences*

A number of social preference models (see Box 3) are based on the assumption that people care about the distribution of payoffs between themselves and a set of relevant reference agents (Fehr and Schmidt 1999; Bolton and Ockenfels 2000; Andreoni and Miller 2002; Charness and Rabin 2002; Fisman, Kariv, and Markovits 2007). These models are a natural starting point for the modelling of social preferences because the behavioral patterns observed in the paradigmatic experiments discussed above strongly suggest that some sort

Box 3. Models of Distributional Preferences

Let π_i and π_j represent the material payoffs of players i and j , respectively. In the *Fehr–Schmidt* (1999) model, individuals are assumed to derive disutility from inequitable outcomes. Inequity is formalized as inequality in the experimental games under consideration. This led to the following utility function:

$$(2) \quad U_i = \pi_i - \frac{\bar{\alpha}_i}{n-1} \sum_{j \neq i} \max\{\pi_j - \pi_i, 0\} - \frac{\bar{\beta}_i}{n-1} \sum_{j \neq i} \max\{\pi_i - \pi_j, 0\},$$

where $\bar{\alpha}_i > 0$ measures the disutility from the average disadvantageous inequality, while $\bar{\beta}_i > 0$ measures the disutility from the average advantageous inequality. The Fehr–Schmidt model implies that players care positively for others' payoffs if they are better off than the other player (advantageous inequality) and negatively if they are worse off than the other player (disadvantageous inequality).

The model by *Bolton–Ockenfels* (2000) stipulates a utility function

$$(3) \quad U_i = f_i(\pi_i, \sigma_i),$$

that is, a player's utility is a function f_i that depends positively on the player's own material payoff and on the relative share, σ_i , that the player receives in the game. If the total payoff in the game, $\Pi = \sum_{j=1}^n \pi_j$, is positive, the relative share is given by $\sigma_i = \pi_i/\Pi$, but if the total payoff is zero, σ_i is assumed to be equal to $1/n$. The model assumes that individuals derive additional utility from a higher σ_i if $\sigma_i < 1/n$, while if $\sigma_i > 1/n$, a higher σ_i is utility decreasing. In the Bolton–Ockenfels model, players do not care about the payoffs of specific other players as long as their “own” relative share, σ_i , is unaffected. In the two-player case, individuals care positively for the other's payoff if they are better off and negatively if they are worse off than the other player.

The distributional preferences in *Charness and Rabin* (2002) are assumed to be given by

$$(4) \quad \begin{aligned} U_i &= (1 - \lambda_i)\pi_i + \lambda_i W(\pi_1, \pi_2, \dots, \pi_n) \\ &= (1 - \lambda_i)\pi_i + \lambda_i \left[\delta_i \min\{\pi_1, \pi_2, \dots, \pi_n\} + (1 - \delta_i)\Pi \right]. \end{aligned}$$

Here, subjects value a *social welfare function*, $W(\pi_1, \pi_2, \dots, \pi_n)$, positively with weight $\lambda_i \in [0, 1]$, while a subject's own payoff, π_i , has weight $(1 - \lambda_i)$. $\lambda_i > 0$ indicates “other-regardingness”, i.e., that player i cares in some way for others' payoffs. The social welfare function has two arguments: the payoff of the least well-off player enters with weight $\delta_i \in (0, 1)$, and the sum of the payoffs, Π , enters W with weight $(1 - \delta_i)$. Thus, if player i cares for others' welfare ($\lambda_i > 0$) and player i cares for the total payoff ($0 < \delta_i < 1$), player i values the payoff of *each* other player positively *regardless of whether player i has a higher or a lower payoff than the other player*. The Charness–Rabin model therefore predicts altruistic behavior if player i cares for the total payoff, and this tendency is particularly pronounced toward the player with the smallest payoff.

Altruistic constant elasticity of substitution (CES) preferences, as in *Andreoni and Miller* (2002) and *Fisman, Kariv and Markovits* (2007), take the form

$$(5) \quad U_i(\pi_i, \pi_j) = \left[(1 - \alpha') \pi_i^\rho + \alpha' \pi_j^\rho \right]^{1/\rho}$$

where $\alpha' \in [0, 1]$ represents the weight on the other player's payoff, while ρ , which obeys $-\infty < \rho \leq 1$, captures the trade-off between equity and efficiency. Altruistic CES preferences and Charness–Rabin type distributional preferences share the property that the payoff of other individuals is never valued negatively.

of interdependent preferences are at play. Another main motivation for the construction of some of these models (Fehr and Schmidt 1999; Bolton and Ockenfels 2000) was the existence of important facts from (i) market experiments with proposer and responder competition and final periods of public good experiments that appeared to contradict the existence—or at least the widespread relevance—of social preferences, and (ii) the observation that the same people are willing to engage in seemingly contradictory behaviors by increasing another's payoff at a cost to themselves in some situations, while decreasing the other's payoff in other situations. For example, positive transfers in the dictator experiment indicate that the dictators value the recipients' payoff positively, while rejections in the ultimatum experiment reduce the proposer's payoff and thus indicate a negative evaluation of the proposer's payoff. Likewise, a rise in the effort level in response to a higher wage leads to an increase in the employer's payoff in the gift exchange experiment, while the third party's sanctions in the third-party punishment experiment reduce the dictator's payoff in that experiment.

Thus, the question is whether social preference models can account for these seemingly contradictory facts. Obviously, a simple model of altruism that assumes that other-regarding individuals value others' payoffs positively cannot account for these facts, nor can a model of spiteful/envious preferences where individuals value others' payoff negatively do so. Since simple models of altruism and spite cannot explain these facts, the concept of a social value orientation (SVO) that social psychologists developed (Liebrand 1984; Liebrand and McClintock 1988; Van Lange et al. 1997) is not capable of doing so because the SVO concept views an other-regarding individual as *either* altruistic/cooperative *or* as envious.

Surprisingly, however, relatively simple models of other-regarding preferences over payoff-distributions—such as those by Fehr and Schmidt (1999) and Bolton and Ockenfels (2000)—go a long way toward reconciling these facts. While this does not mean that these models are necessarily empirically correct, because this requires further empirical testing, it means that widespread seemingly selfish *behaviors* in certain strategic experiments can be perfectly consistent with the existence of widespread *social preferences*.

In the two-player case, a simple, linearized, version of player i 's distributional social preferences $U_i(\pi_i, \pi_j)$ can be written as

$$(1a) \quad U_i = (1 - \alpha)\pi_i + \alpha\pi_j \quad \text{if } \pi_i < \pi_j$$

$$(1b) \quad U_i = (1 - \beta)\pi_i + \beta\pi_j \quad \text{if } \pi_i \geq \pi_j,$$

where both α and β are from the interval $(-1, +1)$; π_i and π_j represent the material payoffs of players i and j , respectively; α is the weight on player j 's payoff when i is behind ($\pi_i < \pi_j$); while β is the weight when i is ahead ($\pi_i > \pi_j$). For simplicity, we drop the subscript i for the weights that player i assigns to the other player's payoff.

Depending on the parameters α and β , this model captures a number of distributional social preference motives discussed in the literature. If $\alpha = \beta = 0$, players are selfish and their indifference curves in the (π_i, π_j) space are vertical (see Figure 1, panel A). If both α and β are negative, a player has *competitive or spiteful* social preferences because he or she always values the other player's payoff negatively.

The indifference curves of *spiteful* players in (π_i, π_j) space are illustrated in Figure 1, panel B. Equations (1a) and (1b) can also capture the *inequality averse preferences* described in Fehr and Schmidt (1999), where subjects dislike both being ahead and being behind. This

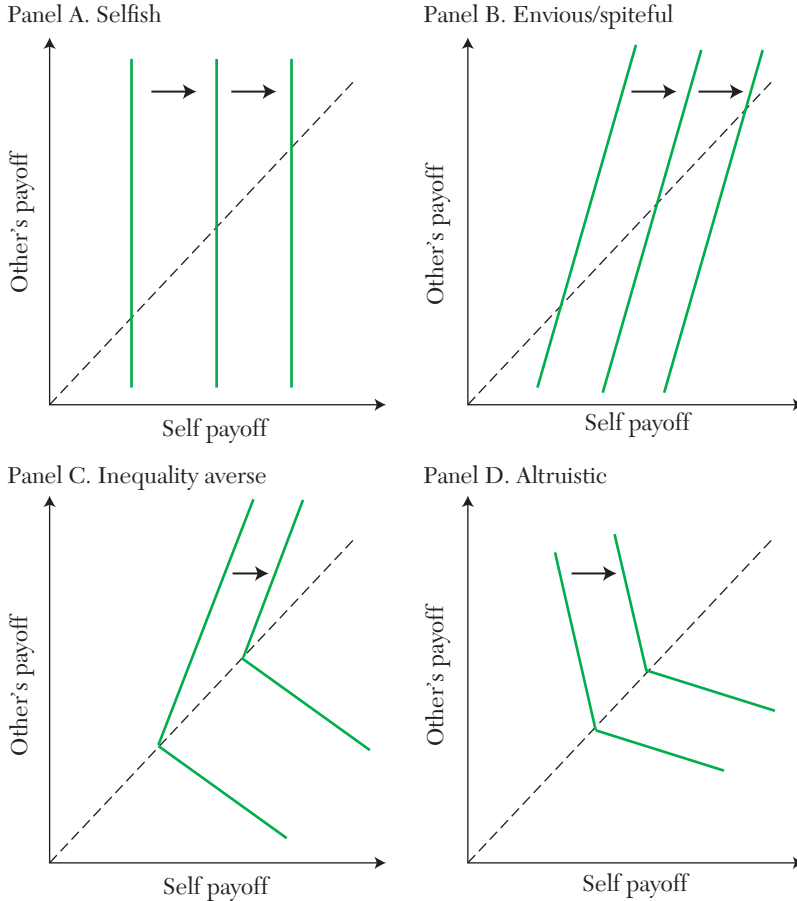


Figure 1. Indifference Curves for Different Types of Distributional Preferences

Notes: The figure shows the indifference curves for different types of distributional preferences captured by equations (1a) and (1b). A: selfish preferences ($\alpha = \beta = 0$). B: envious/spiteful preferences ($-1 < \alpha < 0$, $-1 < \beta < 0$). C: inequality averse preferences ($-1 < \alpha < 0$, $0 < \beta < 1$). D: altruistic preferences ($0 < \alpha < 1$, $0 < \beta < 1$). The slope of the indifference curve in the domain of disadvantageous payoffs is given by $(\alpha - 1)/\alpha$, while the slope in the domain of advantageous payoffs is given by $(\beta - 1)/\beta$.

preference implies that subjects value the other player's payoff positively if ahead but negatively when behind, which can be captured in terms of the parameters in equations (1a) and (1b) by $\alpha < 0$ and $\beta > 0$.² The indifference curves of inequality averse players are illustrated in

²To show this more explicitly: In the two-player case inequality aversion is defined by Fehr and Schmidt as $U_i = \pi_i - \bar{\alpha}_i(\pi_j - \pi_i)$ if $\pi_j > \pi_i$ and $U_i = \pi_i - \bar{\beta}_i(\pi_i - \pi_j)$ if $\pi_i > \pi_j$ with both $\bar{\alpha}_i > 0$ and $0 < \bar{\beta}_i < 1$, implying that players dislike inequality. These utilities can be rewritten as $U_i = (1 + \bar{\alpha}_i)\pi_i - \bar{\alpha}_i\pi_j$ if $\pi_j > \pi_i$ and $U_i = (1 - \bar{\beta}_i)\pi_i + \bar{\beta}_i\pi_j$ if $\pi_i > \pi_j$. Define $\alpha = -\bar{\alpha}_i$ and $\beta \equiv \bar{\beta}_i$ to arrive at equations (1a) and (1b).

Figure 1, panel C. The key property of these indifference curves is that individuals are willing to sacrifice resources to reduce the other player's payoff in order to diminish disadvantageous inequality. In the two-player case, the social preferences Bolton–Ockenfels (2000) assume are quite similar to those of Fehr and Schmidt, that is, the players basically dislike being either behind or ahead.³ Therefore, if a player is behind, she values the other player's preferences negatively, but she values them positively when ahead.

It is easy to see how social preferences like those in Fehr and Schmidt (1999) and Bolton–Ockenfels (2000) can explain that people sometimes increase and sometimes decrease other agents' payoffs. In the Fehr–Schmidt model, the criterion for when individuals switch from being benevolent to malevolent is whether they have more or less than the other's material payoff. As Fehr and Schmidt (1999) point out, and as discussed in depth in Section 3.2, this criterion may not always be the empirically relevant one because equity is not always identical to equality, but it may nevertheless be a useful criterion in many lab and field environments. In the Bolton–Ockenfels model, the criterion for switching from benevolent to malevolent is whether the player receives an equal share of the overall surplus.

How can these theories explain why other-regarding individuals behave very selfishly in competitive market experiments? A simple example may suffice to illustrate this. Consider a population of players consisting only of people with a relatively strong aversion to unequal payoffs in the sense of Fehr–Schmidt (e.g., assume they reject every offer in a *bilateral* ultimatum experiment that is below 40 percent of the available pie). Now put these people into the responder position of a market experiment with *two competing responders* who face a very unfair offer of, say, 3 percent of the pie. The rules of the game are such that if one responder accepts and the other rejects, the accepting responder gets the 3 percent and the proposer gets 97 percent. If both accept, the responder who receives the 3 percent will be randomly drawn and if both reject, all three players receive nothing.

If our fair-minded responders believe that the other responder will accept, they both believe that they will be unable to prevent a very unequal distribution of income. Given these beliefs, they essentially face two very unattractive options in which the proposer always gets 97 percent of the pie and one of the responders gets 3 percent. Therefore, even strongly inequality averse responders will accept the 3 percent offer (proposition 3 in Fehr–Schmidt 1999). Fischbacher, Fong, and Fehr (2009) provide empirical evidence that these belief mechanisms are a key driver of responder behavior in experiments with responder competition. Dufwenberg et al. (2011) show more generally that social preferences do not affect individuals' demand behavior in a perfectly competitive economy if the utility function is separable between an individual's own consumption and the consumption vectors and budget sets of others.

The example above illustrates that the mere belief that other players *behave* selfishly might sometimes induce fair-minded players to do so as well. More generally, perhaps the most important message from papers such as those of Fehr and Schmidt and Bolton and Ockenfels is that the heterogeneity of social preferences is key because—depending on the game played—the existence of selfish players can induce fair-minded players to behave as if they were

³In the two-player case, a version of the Bolton and Ockenfels preferences can be written as $U_i = \pi_i + f(\sigma)$, where $\sigma = \pi_i / (\pi_i + \pi_j)$ measures A's relative payoff. $f(\sigma)$ reflects the other-regarding part of an individual's utility function; it is increasing in σ for $\sigma < 1/2$ (i.e., $\pi_i < \pi_j$) and decreasing for $\sigma > 1/2$ (i.e., $\pi_i > \pi_j$). Note that the preferences in Figure 1, panel C represent a piecewise linear approximation of Bolton–Ockenfels preferences.

selfish and the existence of fair-minded players can induce selfish players to behave as if they were fair-minded.⁴

The simple linear social preference model described in equations (1a) and (1b) can also capture the distributional preferences assumed in Charness and Rabin (2002) that are described in more detail in Box 3. In the two-player case, the weights given to the other player's payoff in the disadvantageous and the advantageous domain, respectively, are given by⁵

$$(6a) \quad \alpha = \frac{\lambda(1 - \delta)}{1 + \lambda(1 - \delta)}$$

$$(6b) \quad \beta = \frac{\lambda}{1 + \lambda(1 - \delta)}.$$

Thus, if i cares about welfare ($\lambda > 0$) and about the total payoff ($0 < \delta < 1$), she values the other's payoff positively even in the disadvantageous payoff domain and the valuation is even higher in the advantageous domain. Note, however, that the only reason why a Charness–Rabin player cares for the other player's payoff in the disadvantageous domain is a concern for the total payoff. Figure 1, panel D illustrates the indifference curves this model implies.

From a behavioral viewpoint, the Charness–Rabin preferences are qualitatively quite similar in the two-person case to nonlinear other-regarding preferences as captured by a CES utility function (see Box 3). Andreoni and Miller (2002) and Fisman, Kariv, and Markovits (2007) have assumed CES preferences. However, while the CES utility function is capable of capturing very similar behaviors, the Charness–Rabin model provides a psychological foundation for altruistic behaviors in terms of a motive to increase the total surplus and to help the worst-off player. Knowing that these motives may underlie other-regarding behaviors can provide a deeper understanding of these behaviors.

As in Charness and Rabin (2002), subjects with these CES preferences always value others' payoffs positively and preferences are linear and given by $U_i(\pi_i, \pi_j) = (1 - \alpha)\pi_i + \alpha\pi_j$ in the special case of $\rho = 1$. It is well known that preferences are Cobb–Douglas for the case of $\rho = 0$, while social preferences take the Rawlsian form $U_i(\pi_i, \pi_j) = \min\{\pi_i, \pi_j\}$ for $\rho \rightarrow -\infty$. In the latter case, indifference curves in the (π_i, π_j) -space become horizontal for $\pi_i > \pi_j$ and vertical for $\pi_i < \pi_j$ with a kink at the 45 degree line, implying that subjects' social preferences are strictly egalitarian. These egalitarian preferences arise in Charness and Rabin for the two-player case if subjects completely disregard their own payoffs (i.e., $\lambda = 1$) and the sum of payoffs (i.e., $\delta = 1$), but care only for the payoff of the least well-off player.

As in Charness and Rabin, CES preferences can thus capture a high willingness to pay to increase the other's payoff when ahead and a very low willingness to pay when behind.

⁴Selfish proposers in the ultimatum experiment have an incentive to make relatively fair offers if fair-minded responders reject low offers. Selfish “employers” in the gift-exchange experiment have a reason to make fair wage offers if fair-minded workers respond to fair offers with higher effort. Selfish players in public goods experiments with punishment have an incentive to cooperate if cooperative individuals punish defectors. In all cases, the existence of players with social preferences can induce fundamental changes in the selfish players' incentives.

⁵In the case where i earns more than j , utility is given by $U_i = (1 - \lambda)\pi_i + \lambda[\delta\pi_j + (1 - \delta)(\pi_i + \pi_j)]$, while $U_i = (1 - \lambda)\pi_i + \lambda[\delta\pi_i + (1 - \delta)(\pi_i + \pi_j)]$ represents the utility in the case where j earns more than i . Reformulating the first function in terms of the weight α on the other player's payoff and the second function in terms of the weight β on the other's payoff yields the expressions in equations (6a) and (6b), respectively.

However, similar to Charness and Rabin's distributional preferences, CES preferences cannot capture subjects' willingness to reduce others' payoffs at a cost to themselves because the other player's payoff is never valued negatively.

3.1.2. Empirical Frequency of Different Distributional Preferences

In the following, we will discuss the empirical properties of distributional preferences that emerged from more systematic attempts to identify the relevant parameters. Fehr and Schmidt (1999) and Bolton and Ockenfels (2000) were inspired by the evidence from the paradigmatic experiments. Their papers provided a unifying account for important, and seemingly contradictory, empirical regularities but they did not involve a systematic attempt to identify the model parameters empirically. To achieve identification of the parameters of distributional preferences like those in equations (1a) and (1b), one needs information about subjects' behavior on many budget lines with different slopes in the (π_i, π_j) space. In other words, one needs a large set of generalized dictator experiments in which one subject can unilaterally determine the distribution of payoffs between herself and another subject at various costs of redistributing payoffs. By systematically varying the slope of negatively sloped and positively sloped budget lines, it is possible to identify individuals' indifference curves in the domain of advantageous and disadvantageous inequality and to determine which of the fundamentally different distributional preference types illustrated in Figure 1 describes the behavior of (sub) groups or individuals best.⁶ In addition, this approach also enables a clean separation of truly selfish individuals from other-regarding individuals because it allows for deviations from selfishness when the costs of other-regarding acts are low.⁷

Concern for Equality or Concern for the Total Payoff?—Charness and Rabin (2002) took an important step toward answering this question in a study with roughly 220 undergraduate students in 29 distinct experimental games or allocation tasks. In one of their tasks, 69 percent of the participants chose (Own, Self) payoffs of (400, 750) rather than (400, 400). This result, and similar other results (Charness and Grosskopf 2001) imply that roughly two thirds of these student populations are unwilling to reduce others' income for the sake of equality, implying that $\alpha \geq 0$. However, these results do not imply that subjects have a positive concern for the total payoff because it costs nothing to choose the allocation that maximizes the total payoff.

⁶There is a considerable literature on social value orientation (SVO) in psychology that uses the so-called ring measure of SVO (Liebrand 1984; Liebrand and McClintock 1988) and/or the triple-dominance measure of SVO (Van Lange et al. 1997; Van Lange 1999) or the slider task (Murphy, Ackermann, and Handgraaf 2011; Murphy and Ackermann 2014). These measures are based on generalized dictator experiments, but they do not lend themselves easily to the estimation of utility functions. For example, they do not allow us to differentiate between the four qualitatively different types of indifference curves displayed in Figure 1. Instead, they use measures such as the ratio between the total payoff given to the other player and the total payoff assigned to "self" (across all dictator experiments) to assign individuals to predefined SVO types such as "cooperative" (= desire to maximize joint gains), "altruistic" (= desire to maximize the other player's payoff), and "competitive" (= desire to maximize the payoff difference). In particular, the SVO measures cannot identify inequality aversion as defined in Fehr and Schmidt (1999) or Bolton and Ockenfels (2000). For this reason, we do not use them in the following review of distributional preferences.

⁷Some people may be inclined to discount situations in which the cost of other-regarding acts is low, but social life is in fact pervaded by situations in which low-cost favors and low-cost punishments can be given to other people. When a colleague in the workplace asks for help, when a stranger in a city asks for directions, or when a colleague punishes you for your performance with a dismissive smile, the costs for the helper or punisher are often very low, while the benefits and costs for the receiving party are high.

Therefore, choosing the allocation that maximizes the total payoff in this task is compatible with selfishness (i.e., with $\alpha = 0$).

A positive concern for the total payoff is, however, indicated in an allocation task from Charness and Rabin (2002) where 50 percent of the subjects choose (375, 750) rather than (400, 400), implying that half of the students are willing to pay \$1 to increase the other's payoff by \$14. Based on these results, Charness and Rabin conclude that "inequality reduction is not a good explanation of Pareto-damaging behavior" (p. 819).

The paper by Charness and Rabin led to a lively discussion of the relevance of different motives underlying distributional preferences. Engelmann and Strobel (2004), who conducted experiments with roughly 380 undergraduate economics and business administration students, concluded that inequality aversion is only relevant to the extent to which it coincides with the desire to help the least well-off individuals, while the desire to increase the whole group's payoff is a robust feature of distributional preferences. The generality of their conclusions has been challenged by studies showing that noneconomists put a substantially higher weight on equality—compared to the maximization of the group's overall payoff—than economists do (Fehr, Naef, and Schmidt 2006).

Many other studies have, in the meantime, estimated the structural parameters α and β of model (1) above based on observations from strategic and nonstrategic experiments. A recent meta-analysis by Nunnari and Pozzi (2022) uses data from 41 articles, comprising student and nonstudent populations, with a total of 297 estimates of the different populations' *average* α and β values. The authors estimate the overall distribution of parameters across populations and find weighted mean values for $\alpha = -0.533$ (standard error of 0.110) and $\beta = 0.326$ (standard error of 0.036). This indicates that across all the (strategic and nonstrategic) experiments in the meta-analysis, the observed behaviors are consistent with inequality aversion for the representative agent. A similar qualitative conclusion is reached if one restricts attention to the dictator experiments, which arguably offer the cleanest preference interpretation of the estimated parameters: the mean values of α and β are then given by $\alpha = -0.266$ (standard error of 0.096) and $\beta = 0.387$ (standard error of 0.038), again indicating that subjects display inequality aversion *on average*. In contrast, in strategic games—where players are more likely to view each other as opponents—disadvantageous inequality aversion is much higher ($\alpha = -0.727$) while advantageous inequality aversion is significantly lower ($\beta = 0.214$).

Preference estimates of the representative agent typically hide important heterogeneity at the individual level. A systematic analysis of individual heterogeneity in distributional preferences has been conducted in the seminal papers by Andreoni and Miller (2002) and Fisman, Kariv, and Markovits (2007), who estimated individuals' CES utility functions (see Box 3). Their results indeed show enormous heterogeneity in subjects' preferences. In addition, the Fisman–Jakiela–Kariv–Markovits group shows striking differences between students and the general population's distributional preferences in a series of papers (e.g. Fisman et al. 2015): the general population is much more other-regarding (i.e., has a much higher α' in equation (5)) and puts a much higher weight on equality (i.e., displays $\rho < 0$ in equation (5)) compared to students. This also holds if one controls for age.⁸ Because of space limitations and because the CES approach neglects important classes of distributional

⁸The large differences between student samples and the broader population are consistent with research reported in Snowberg and Yariv (2021) and Cappelen et al. (2015).

preferences—inequality aversion and envy—we describe the results of these studies in more detail in Supplemental Appendix 1.

*How Prevalent Are Inequality Aversion and Spite?*⁹—Kerschbamer (2015) developed a systematic approach—the equality equivalence test (EET)—that enables the identification of all four distributional preference types displayed in Figure 1. For this purpose, subjects are presented with choice lists in the domain of disadvantageous inequality (DA lists) and the domain of advantageous inequality (A lists). In any given list, the subjects face a series of binary choices where the equal payoff distribution E is always paired with an alternative allocation (See Figure A2 in Supplemental Appendix 2). A list essentially confronts subjects with a series of positively and negatively sloped budget lines in the “self-payoff/other payoff” space with two discrete options on each budget line.

Table 1 shows the results of a relatively large number of EETs in student samples (Kerschbamer 2015; Balafoutas, Kerschbamer, and Sutter 2012; Paetzel, Sausgruber, and Traub 2014; Balafoutas et al. 2014; Krawczyk and Le Lec 2021) and in broad, demographically diverse population samples (Chapman et al. 2023; Kerschbamer and Muller 2020; Hedegaard et al. 2021). Several striking facts emerge from these studies. First, the share of selfish individuals is much larger in the student samples, where it varies between 29 percent and 58 percent, while it varies only between 5 percent and 20 percent in the broad population samples.

Second, the much larger share of other-regarding subjects in the broad population samples is primarily due to a much larger share of inequality averse individuals in these samples. While the share of inequality averse individuals in student samples varies between 7 percent and 12 percent, this share is between 23 percent and 68 percent in the broad population samples. Third, the share of envious subjects is also slightly higher in the broad population samples, where it varies between 8 percent and 14 percent, while in the student samples it varies between 2 percent and 10 percent.

Finally, the share of individuals with altruistic preferences is considerably larger in student samples than in broad population samples. The average share of altruistic subjects over samples consisting only of students amounts to 37 percent, while the average share of altruists is only 19 percent in the broad samples. Taken together, the general population seems much more other-regarding and this higher share of other-regarding individuals reflects a much larger share of inequality averse subjects and a somewhat larger share of envious subjects. Interestingly, however, altruism is more prevalent in student samples.⁹

The existence of large differences between the student samples and the general population samples are corroborated by two earlier studies by Bellemare, Kröger, and van Soest (2008, 2011). Using a representative sample of the Dutch population that played ultimatum and dictator experiments, they estimated a nonlinear version of the Fehr–Schmidt model and find that the young and educated individuals in their sample display a substantially lower aversion to disadvantageous inequality than the rest of the sample.

The main findings from the EETs are further corroborated by studies that identify preference clusters without committing to a prespecified number of preference types (Fehr®Epper®Senn 2023a; Epper et al. 2020; Henkel® et al. 2024). In these studies, a

⁹When comparing student samples and broad population samples one may worry about stake size effects because in the general population income variation is much larger. However, the prevailing evidence (e.g., in Epper®Senn®Fehr 2024) indicates that income levels do not affect the estimates of α and β .

TABLE 1
EMPIRICAL FREQUENCY OF DIFFERENT DISTRIBUTIONAL PREFERENCE TYPES

Study	Subject pool	Preference types in the EET			
		Altruistic	Inequality averse	Envy/spite	Selfish
Kerschbamer (2015)	$N = 92$ Austria	33.7%	11.9 %	3.2%	48.9%
Krawczyk and Le Lec (2021)	$N = 101$ Poland	48.5%	11.9%	9.8%	28.7%
Balafoutas, Kerschbamer, and Sutter (2012)	Student samples $N = 132$ Austria	28%	8.3%	5.3%	58.3%
Balafoutas et al. (2014)	$N = 195$ Austria	49.7%	6.7%	7.2%	33.8%
Paetzl, Sausgruber, and Traub (2014)	$N = 280$ Germany	29.3%	9.3%	1.8%	58.3%
Hedegard et al. (2021)	$N = 885$ Denmark	47.1%	23.2%	8.6%	20.0%
Kerschbamer and Müller (2020)	Broad population samples $N = 2,794$ Germany	13.4 %	64.8%	14.0%	5.0%
Chapman et al. (2023)	$N = 1,000$ USA	27.5%	41.9%	8.3%	16.6%
2nd wave German Internet Panel ^a	$N = 2,583$ Germany	11.5%	67.8%	11.0%	7.9%
		Preference types in endogenously emerging preference clusters			
		Altruistic	Inequality averse	Predominantly selfish	
Epper et al. (2020)	Broad population samples $N = 3,691$ Denmark	30.2%	37.3%	32.5%	
Fehr®Epper® Senn (2023b)	$N = 816$ Switzerland 2017	34.4%	50.8%	14.8%	
Henkel®Fehr®Senn ®Epper (2024)	$N = 916$ Switzerland 2020	30.5 %	45.5%	24.0%	

Notes: All studies mentioned in the table are based on incentivized one-shot experiments with anonymous partners. An individual is classified as altruistic in the EET (in terms of the parameters of equation (1a) and (1b)) if her choices imply $\alpha \geq 0$ and $\beta \geq 0$ with at least one strictly positive. To be classified as envious/spiteful, either $\alpha < 0$ and $\beta \leq 0$ or $\alpha \leq 0$ and $\beta < 0$ holds. Individuals are classified as inequality averse in the EET if $\alpha < 0$ and $\beta > 0$ holds. The derivation of endogenous preference clusters in the three final studies is based on the application of a nonparametric Bayesian clustering algorithm (Dirichlet process means). The symbol ® indicates that the author order has been randomly determined.

^aThe results in the Kerschbamer and Müller (2020) paper are based on a first application of the EET to the German Internet Panel (GIP) in 2016. In the meantime, the EET was performed with this sample a second time in 2018. We therefore also include the results of this second wave of data collection to illustrate the stability of the preference classification over time in Table 1.

nonparametric Bayesian clustering algorithm (“Dirichlet Process Means, DP-Means”) is applied to two large Swiss and one Danish data set (see Table 1) where subjects faced many

positively and negatively sloped budget lines that allow for the identification of distributional preferences. The algorithm determines the number and the behavioral properties of the preference clusters endogenously and assigns each individual to one of the clusters.

The clustering algorithm yields the same three (qualitative) preference clusters in all three broad population samples: an altruistic cluster, an inequality averse cluster, and a cluster of individuals who behave predominantly selfishly.¹⁰ Moreover, the inequality averse cluster represents the majority group and the altruistic and inequality averse cluster together comprise between two-thirds and 85 percent of the sample across the different data sets. Thus, the endogenous clustering approach is also in line with the conclusion that inequality aversion and altruism are the dominant patterns of distributional preferences in the general population, while predominantly selfish subjects constitute a smaller yet significant share of the population

3.2 *The Role of Merit and Luck in Distributional Preferences*¹¹

In the standard dictator experiment and other pure distribution experiments, the experimenter exogenously provides the income that can be divided among the parties. Moreover, the role of the powerful player, who can make a unilateral allocation decision, is assigned randomly, and the subjects typically interact anonymously with each other and have no information about the other players' income, wealth, and social background. It seems natural that no player in such an environment has a priori a greater normative claim on the available resources compared to the other players, that is, equality is a natural reference point for judging the fairness and equitability of outcomes. It is, however, also clear that social background, the saliency and status of particular individuals, the social proximity among individuals, as well as their effort and contribution to the available resources, including the risks and conditions under which they had to produce these resources, can play a role in what is considered an equitable distributional claim. The potential importance of these factors is widely acknowledged and may often drive a wedge between equality and equity (fairness) and offers ample opportunities for examining these factors in a controlled way.¹²

In this context, the notions of merit, luck, and effort play a key role. In many modern societies, meritocracy seems a widely held normative ideal. A prominent theory of equality of opportunity (Roemer and Trannoy 2015) argues that individuals should be held responsible for factors under their own control, while they should not be held responsible for factors beyond their control. Based on this approach, there is no normative reason to redistribute resources that are entirely generated by an individual's effort, while earnings that accrue at least partly from noncontrollable external factors ("luck") should be subject to redistribution.

¹⁰The frequency of envious/spiteful subjects is generally small and their location in preference space is scattered; they thus do not constitute a distinct cluster but are subsumed either under the selfish or the inequality averse cluster.

¹¹Due to space constraints, we have moved the discussion of the role of risk in social preferences to Supplemental Appendix 3.

¹²The very fact that equity and equality may diverge induced Fehr and Schmidt to first call the preferences they examine as "*inequity* aversion," i.e., "a general dislike for outcomes that are perceived as inequitable." However, under the special circumstances in the laboratory experiments they examined, where subjects entered the laboratory as equals, knew nothing about each other's background, and were randomly allocated to different roles, they assumed that equality may be a reasonable reference point.

3.2.1 *Equity and Entitlements*

Psychologists and sociologists (Homans 1961; Adams 1963, 1965) have developed positive theories of justice that incorporate a widely applicable merit principle. According to this view, individuals perceive inequity (i.e., unfairness) if their outcome/input ratios diverge from the ratio of their relevant comparison agent. The outcomes and inputs that enter the equity/fairness calculus have typically been interpreted very broadly, that is, any kind of reward that the individuals experience can represent an outcome and any kind of perceived contribution to the outcome can represent an input.

Equity theory thus allows for many subjective influences that may make its predictions rather malleable and imprecise. However, it makes sharper predictions if restricted to more easily measurable outcomes (e.g., received material payoffs) and inputs (e.g., time spent on a task or material payoff contributed to the group payoff). Mikula and Uray (1973) conducted, for example, an experiment with recruits from the Austrian Army to examine whether subjects follow the equity/fairness principle defined above or the equality principle. Two matched parties produced a joint monetary payoff, and then one of the individuals could unilaterally allocate this payoff to the two parties. They found that the parties with poorer performance assigned themselves more than the proportional monetary payoff equity theory predicted but less than the equal monetary payoff. Likewise, the parties with higher performance requested *on average* more than the equal split but less than the reward equity theory predicted. For example, when the performance ratio between the poorer and better performers was between 62.5 percent and 37.5 percent, the higher performers requested on average 54.5 percent of the joint payoff, while the lower performers requested 42.7 percent. These findings are consistent with the view that both equity concerns as defined above and equality concerns affect behavior.

More generally, the data confirm the qualitative equity theory prediction that better performers request (if they have the power to decide) and are conceded (if the partner has the power to decide) a higher share of the joint payoff. Likewise, poorer performers allocate themselves and are allocated a lower share of the joint payoff (Leventhal and Michaels 1969; Leventhal and Anderson 1970; Leventhal and Lane 1970; Lane and Messe 1971; Leventhal and Michaels 1971). This literature thus shows that subjects behave as if they feel entitled to a larger (smaller) share if they contribute more (less) to the joint payoff than their experimental counterpart.

One drawback in these experiments was that participants were typically deceived in various ways. Often, there was no actual working partner who contributed to a joint surplus or the performance ratios to which the parties were randomly allocated did not reflect the subjects' actual performances, or individuals had to make hypothetical choices without real economic consequences. These practices may have generated doubts about the credibility of the implemented procedures or might have affected subjects' behavior in other ways. For this reason, it makes sense to ask whether the notion of "earned entitlements" or "earned property rights" implied by equity theory is indeed robust to the methods used in experimental economics which rule out deception while implementing designs with transparent economic consequences for the involved parties.

The experimental economics literature (e.g., Hoffman et al. 1994; Ruffle 1998; Fahr and Irlenbusch 2000; Konow 2000; Cherry, Frykblom, and Shogren 2002; Frohlich, Oppenheimer, and Kurk 2004; Cappelen et al. 2007; Krawczyk 2010; Lefgren, Sims, and Stoddard 2016) also strongly suggests that "earned entitlement" effects exist. For example, Hoffman et al. (1994) and Cherry, Frykblom, and Shogren (2002) show that the dictators take a higher share of the

pie when they earned the role of the dictator in a quiz or when they generated the pie that can be distributed.¹³ While Hoffman et al. (1994) and Cherry, Frykblom, and Shogren (2002) show that the dictators behaved more selfishly if they acquired an earned entitlement, Konow (2000) and Ruffle (1998) show that the dictators also respect the recipients' earned entitlement, that is, the recipients' earned entitlements also constrain the dictators' selfish behavior. Subjects in Konow's experiment *jointly* produce the pie to be distributed in the subsequent dictator experiment in a (letter production) task that enables the exact measurement of the matched subjects' relative contributions (i.e., the number of letters). He hypothesizes that fairness considerations will induce the dictators to tilt the allocation given to the recipients toward their relative contribution in the production task—a finding that his data nicely corroborates.

3.2.2 Modeling Entitlement Effects

How should we model the motivational forces underlying the earned entitlement effect? To set the stage for this discussion, it is useful to rely on a concrete example that is taken from the seminal paper by Cappelen et al. (2007), who introduced the notion of “pluralistic fairness ideals” to the literature. In their experiment subjects participate first in two production tasks where they can finance an investment, q_i , that generates an output, $x_i = a_i q_i$, from an identical endowment E . Each participant was randomly assigned to a high ($a_i = 4$) or a low ($a_i = 2$) rate of return on investment.

After the two production tasks, each subject was matched twice randomly with a different partner and played a bilateral dictator experiment with each of the two partners. The pie size in each dictator experiment was given by the paired players' jointly produced total output, $X = a_1 q_1 + a_2 q_2$. Note that some pairs in this experiment face a total output produced with identical returns on investment ($a_1 = a_2$) and identical investments ($q_1 = q_2$), while others can distribute an output that is produced with different rates of return ($a_1 \neq a_2$) and/or different investment levels ($q_1 \neq q_2$).

In the following, we discuss the behavioral predictions for this experiment for (i) inequality averse individuals, (ii) individuals motivated by equity theory, and (iii) individuals who are motivated by one of the fairness ideals stipulated in Cappelen et al. (2007). An inequality averse dictator would implement the allocation that equalizes the two players' incomes. If we denote s_1 (s_2) as player 1's (player 2's) share of the total payoff, X , income equalization implies $E - q_1 + s_1 X = E - q_2 + s_2 X$. From this, and the fact that $s_1 + s_2 = 1$, follows that player 1's share is given by

$$(7) \quad s_1^{IA} = \frac{1}{2} + \frac{q_1 - q_2}{2X},$$

where the superscript *IA* refers to the predicted share under inequality aversion. Thus, inequality averse dictators assign individuals with identical investments the same income because they

¹³Hoffman et al. (1994) also claim that if one not only introduces anonymity between the subjects but also anonymity between the subjects and the experimenter, the dictators behave more selfishly and take a higher share. However, subsequent research (Frohlich, Oppenheimer, and Kurk 2004) suggests that the way Hoffman et al. implemented experimenter–subject anonymity also generated doubts among the subjects about whether the recipient was a real person, which may well induce selfish behavior. Other research on double anonymity (e.g., Bolton, Katok, and Zwick 1998; Barmettler, Fehr, and Zehnder 2012) indicates that lack of subject–experimenter anonymity has only minor, insignificant effects in dictator, ultimatum, and trust experiments.

view them as equally meritorious, while individuals who invest more are assigned a higher income share proportional to the investment gap between the players. Inequality aversion therefore incorporates a notion of meritocracy because the principle of equality is applied to *all* material payoffs, and effort or investment costs are certainly part of this. In contrast, according to equation (7), inequality averse individuals do not honor the luck of those participants who were randomly assigned a higher rate of return a_i .

Which allocation would a dictator who is motivated by the forces stipulated by equity theory implement in the Cappelen et al. experiment? He or she would equalize the output/input ratios between the players. While a plausible interpretation of the “output” seems to be the earnings received, there is more ambiguity when considering the inputs. This depends crucially on whether one considers the investment levels, q_i , or the contributions to the total earnings, $x_i = a_i q_i$, as an “input.” However, if we view equity theory as embodying a merit principle, the investment levels are the appropriate input measure, implying that equity is established once $s_1 X/q_1 = s_2 X/q_2$ or $s_1/q_1 = s_2/q_2$ holds.

If we consider that $s_1 + s_2 = 1$, the payoff shares predicted by equity theory, s_i^E , are given by the players’ relative investment shares:

$$(8) \quad s_1^E = \frac{q_1}{q_1 + q_2} \quad \text{and} \quad s_2^E = \frac{q_2}{q_1 + q_2}.$$

This coincides with the notion of fairness advocated by Konow (2000) and with the liberal egalitarianism fairness ideal proposed in Cappelen et al. (2007). Note that the notion of fairness described in (8) also implies that liberal egalitarians do not honor luck, in the form of randomly assigned investment returns, while honoring merit, in the form of players’ relative investment shares.

In addition to the liberal egalitarian fairness ideal, Cappelen et al. introduce two further potentially relevant fairness ideals: the strictly egalitarian ideal (s_i^{SE}) and the libertarian ideal (s_i^L). According to the strictly egalitarian ideal, the earnings should be distributed equally regardless of the players’ investments and rates of return, that is, $s_i^{SE} = 1/2$. In contrast, according to the libertarian ideal each player is entitled to the earnings he or she produced regardless of any luck or effort considerations, that is, $s_i^L = a_i q_i/X$.

What kind of utility function rationalizes the behavior of players who are motivated by a fairness ideal? To answer this question let s_i^F be the income share of i under a generic fairness ideal, $s_i^F \in \{s_i^E, s_i^{SE}, s_i^L, s_i^{IA}\}$. A plausible utility function that incorporates the players’ fairness ideals could look similar to the Fehr and Schmidt utility function, where the reference point is no longer the other players’ payoff but the dictator’s own fair share as defined by s_i^F , and with actual income denoted by y_i :

$$(9) \quad U_i = y_i - \bar{\alpha}_i \max\{s_i^F X - y_i, 0\} - \bar{\beta}_i \max\{y_i - s_i^F X, 0\}.$$

A dictator with this utility function would never give herself less than $s_i^F X$ and would allocate herself exactly $s_i^F X$ if $\bar{\beta}_i > 1$. In contrast, Cappelen et al. (2007) assumed that deviations from the fair payoff impose symmetric (quadratic) costs on subjects, which led them to postulate the utility function:

$$(10) \quad U_i = y_i - \gamma_i \frac{(y_i - s_i^F X)^2}{2X}.$$

The two utility functions above imply that there is a nonpecuniary disutility from receiving an inequitably large share in case of $y_i > s_i^F X$. Thus, because the players' income shares add up to one, subjects in this situation are willing to reduce their own incomes and behave generously toward the other player. Likewise, the associated nonpecuniary disutility from receiving a too-low share generates a willingness to pay to reduce the other player's payoff in case of $y_i < s_i^F X$.

To what extent are the different fairness ideals capable of capturing earned entitlement effects? Liberal egalitarians and inequality averse individuals who behave according to utility functions (9) or (10) will exhibit earned entitlement effects if they contributed more effort or investments to the production process. In contrast, libertarians with a fairness ideal of $s_i^L = a_i q_i / X$ will show an earned entitlement effect even if they provide higher output merely because of luck (i.e., a higher a_i).

What are the empirical results of the Cappelen et al. (2007) study? Based on the assumption that there are three types of fairness ideals— s_i^E , s_i^{SE} and s_i^L —and that the subjects behave according to utility function (10), the authors estimate a mixture model that provides the relative share of individuals who are assigned to the different fairness ideals and also an estimate of the distribution of the strength of fairness preferences as captured by γ . They find that the share of strict egalitarians is 43.5 percent, the share of liberal egalitarians is 38.1 percent, and the share of libertarians is 18.4 percent. In another study with a broader sample of Norwegian and German students (Cappelen et al. 2013), the allocation of individuals to the different fairness ideals is different, however, as only 22.5 percent of individuals are strict egalitarians, while 42.5 percent are liberal egalitarians, and 35 percent are libertarians.

The introduction of heterogeneous fairness ideals by Cappelen et al (2007) represents an interesting innovation; it implies that if people do not share in one particular situation, then this may not mean that they do not care about fairness (i.e., put no weight on fairness in (9) or (10)), but that they consider sharing unfair. Many interpretations of experiments overlook this distinction and too often categorize people who do not share as selfish. The distinction between the weight put on fairness and the concrete fairness ideals also opens an avenue toward studying the impact of institutions on distributional preferences. It has been argued, for example, that studying economics makes people more selfish (Frank, Gilovich, and Regan 1993). An interesting question in this context is whether studying economics indeed reduces the other-regarding component in individuals' utility function (i.e., $\bar{\alpha}_i$, $\bar{\beta}_i$, or γ_i in equations (9) and (10)) or whether it changes people's fairness ideals. More generally, fairness ideals are likely to be shaped by educational and political institutions as well as by the general moral infrastructure of societies.

The Cappelen et al. study deserves credit for introducing and estimating the share of subjects assigned to *heterogenous* fairness ideals that are based on different normative recognitions of "luck" and "effort." At the same time, it is clear that the assumed existence of three exogenously given fairness ideals is a strong assumption, and that it would be desirable to develop experimental designs and econometric methods that make it possible to infer individual subjects' normative reference points from the data, instead of exogenously assuming them.

Likewise, while assuming that different reference points can capture the distinction between luck and effort, it is also possible that individuals' fairness ideals directly affect their interpretation of an individual's deservingness. For example, an individual with a meritocratic fairness ideal who faces a "lazy" individual may put a different weight on being fair toward this individual compared to a situation where she faces a hard-working individual. Thus, we believe it is an

important task for future research to find ways to *simultaneously* identify normative reference points *and* the strength of fairness preferences (in terms of $\bar{\alpha}_i$, $\bar{\beta}_i$ or γ_i) at the individual level, and to determine how different environments affect these parameters.¹⁴

Another important unresolved question is how do empirical measures of inequality aversion, altruism, and selfishness, which are identified in experiments without obvious entitlements (like, e.g., in Table 1), relate to the different fairness ideals? For example, are subjects identified as inequality averse in Table 1 liberal egalitarians or strict egalitarians? Likewise, how are subjects identified as selfish in experiments without obvious entitlements allocated to the different fairness ideals? These questions are largely unexplored, but a first study by Fehr@Epper@Senn (2023a) shows that individuals who are identified as selfish in experiments without obvious entitlements also behave very selfishly in experiments with entitlements, that is, fairness ideals do not seem to matter to them. In contrast, individuals classified as inequality averse or altruistic in experiments without obvious entitlements display considerable concern for meritocracy. 70 percent of the altruists (60 percent of the inequality averse individuals) displayed a concern for meritocracy by, e. g., conceding a higher payoff to a better-performing player, while 30 percent of the altruists (40 percent of the inequality averse) chose always the strictly egalitarian allocation.

This finding confirms that meritocratic concerns only motivate other-regarding individuals, and it has implications for interpreting the relevance of impartial spectator experiments that are frequently used to identify individuals' fairness ideals. In these experiments, an impartial spectator typically allocates payoffs between two *other* players, but the allocation decision does not affect the spectator's payoff. Therefore, one cannot identify the selfish individuals for whom fairness ideals have little behavioral bite when their self-interest is at stake. The assignment of fairness ideals based on impartial spectator experiments may thus run the danger of overstating the behavioral relevance of fairness ideals because if $\bar{\alpha}$, $\bar{\beta}$, or γ in equation (9) or (10) are zero, fairness ideals are behaviorally irrelevant.

3.2.3 Cultural Differences and the Relative Importance of Fairness and Efficiency Concerns

Most experiments documenting entitlement effects have been conducted with Western student populations. These subjects are part of an environment that permanently evaluates their performance and requires high effort levels to pass frequent examinations, provide satisfactory work results, or move up the career ladder. Meritocratic and libertarian ideas may well flourish in this environment, raising the question about the prevalence of difference fairness ideals in the broader population and in other cultures.

Jakiela (2015) studies entitlement effects in a younger population in rural villages in Kenya and compares them with those observed in a US student population. Jakiela's experiment is motivated by observations that these entitlement effects may not exist in poor rural communities with strong traditions of solidarity and mutual assistance (Platteau 2000). She implemented so-called taking treatments, where a dictator could allocate money that

¹⁴Cabeza (2021) conducted a study that varied the deservingness of players along the "effort" and "luck" dimensions. She reports that changes in deservingness are associated with changes in the strength of fairness preferences in both the domains of advantageous and disadvantageous inequality. If subjects become "more deserving," the decision-makers' willingness to behave altruistically in the domain of advantageous inequality increases while the willingness to behave enviously decreases.

the recipient won or earned. There was a luck condition, where the recipient's roll of a die determines the size of the overall budget, and an effort condition, where the recipient produced the budget in a real effort task. The question then is whether the dictators in this experiment respect the "earned entitlements" in the effort condition by allocating a higher share to the recipients compared to the luck condition. However, while US students showed strong behavioral differences between the luck and effort conditions, there is no evidence that the Kenyan village population respected the recipients' earned entitlements in the effort condition. In fact, the recipients received even a lower share of the pie in the effort compared to the luck condition.

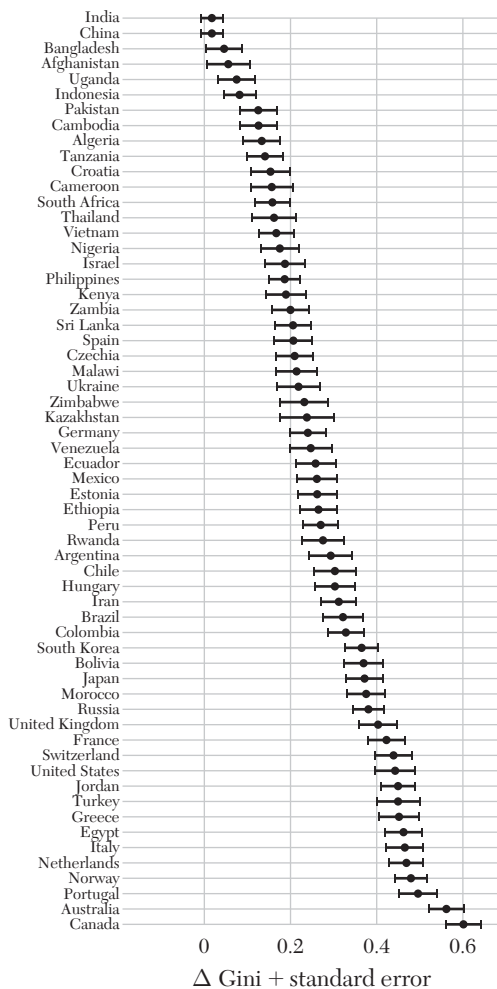
Almås, Cappelen, and Tungodden (2020) and Almås et al. (2024; 2025) were the first to study the relevance of the distinction between "luck" and "effort/merit" in *broad* population samples. In Almås et al. (2024, 2025), they conducted a large study with 65,800 participants that comprise representative country samples taken from 60 different countries around the globe. In each country, at least 1,000 individuals participated in the experiment as impartial spectators who had the option to redistribute income between two subjects ("workers"). Initially, before redistribution, one of the workers had an income equivalent to US \$6 while the other had US \$0. In the luck treatment, a lottery draw (from an earlier experiment on Mechanical Turk in which the workers participated) generated the inequality between the subjects. In contrast, the productivity differences between the two workers in a real effort task determined the initial inequality in the merit treatment. The impartial spectator could redistribute the \$6 in \$1 units without any cost in both the luck and the merit treatments.

In addition to the luck and merit treatments, the authors also conducted a so-called efficiency treatment. This treatment is identical to the luck treatment except that redistribution is associated with large cost: for every \$1 given to the poorer workers, the income of the worker who was initially richer is reduced by \$2. With such a large redistribution cost, one would expect a substantial increase in inequality acceptance. This study enables the examination of the potential universality of the role of merit versus luck versus efficiency costs on preferences for redistribution in a highly controlled experimental set-up that is kept constant across all countries.

Several findings in Almås et al. (2024, 2025) stand out (see Figure 2). First, there are huge cultural differences in the extent to which the merit treatment causes an increase in inequality acceptance relative to the luck treatment (Figure 2, panel A). The merit treatment causes basically no increase in inequality in countries like India and China, presumably because the implemented inequality is already very high in the luck treatment. However, in the overall sample, the Gini coefficient in the merit treatment is 26 percentage points higher than in the luck treatment, and the implemented Gini coefficient increases by 50–60 percentage points in the merit versus the luck treatment in countries like Canada, Australia, or Portugal.

Second, the effect of the efficiency treatment, when compared to the luck treatment, is much lower than the effect of the merit treatment in almost all countries (Figure 2, panel B and panel C). On average, the Gini coefficient in the efficiency treatment is only 5 percentage points higher than in the luck treatment, and the treatment effect is zero or even negative in many countries. The efficiency treatment increases inequality acceptance at most by (slightly less than) 20 percentage points. Almås et al. (2024, 2025) thus document the near universality of the important role of merit in redistributive preferences, including the near universality of the greater role of fairness considerations relative to efficiency considerations for redistribution.

Panel A. Treatment effect merit



Panel B. Treatment effect efficiency

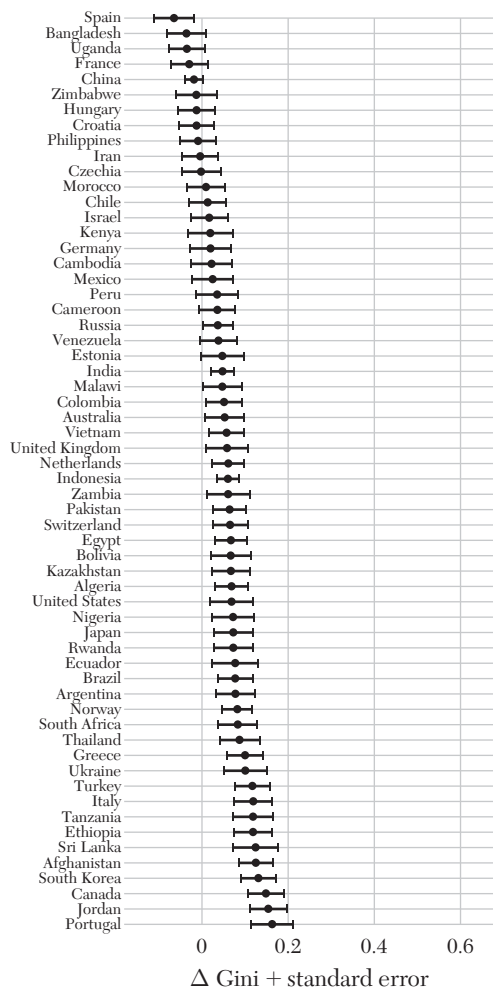


Figure 2. The Relative Importance of Merit and Luck for Distributional Preferences

(continued)

Third, the study finds vast differences in fairness ideals across countries. While almost 75 percent of the subjects display a libertarian ideal in countries like India and China, this share is in the range of 10–15 percent in countries like Canada, Australia, or Norway. Conversely, the share of meritocratic ideals is vanishingly small in India and China, while it is close to 50 percent in Canada, Australia, and Norway. Interestingly, the cultural/country differences in fairness ideals primarily show up in the different shares of libertarians versus meritocrats, while the share of strict egalitarians is roughly between 20 and 30 percent in most countries.

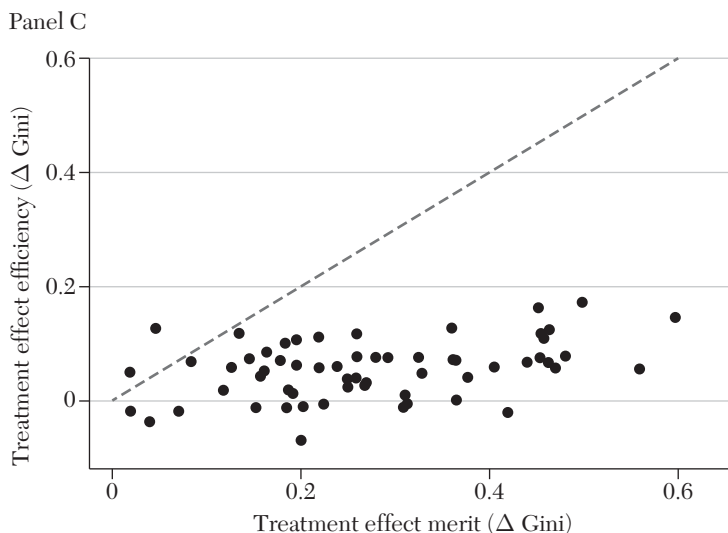


Figure 2. The Relative Importance of Merit and Luck for Distributional Preferences (*continued*)

Notes: The figure is based on data from Almås et al. (2025). It shows how the introduction of performance-dependent inequality in the merit treatment (Figure 2, panel A) and costly redistribution in the efficiency treatment (Figure 2, panel B) increases the implemented inequality (Δ Gini) relative to the luck treatment for each country. The treatment effects are estimated with a country-specific regression controlling for prespecified background characteristics. Figure 2, panel C indicates that merit concerns are much more influential than efficiency concerns.

3.2.4 Shallow Meritocracy?

The distinction between earnings generated through individuals' choices and earnings that accrue to individuals because of lucky circumstances is at the heart of the meritocratic fairness ideal. A key issue in this concept of fairness is, however, that choices are endogenous. In other words, individuals' choices are themselves affected by circumstances. For example, an individual who faces discrimination will generally have weaker incentives to exert high effort because the reward for effort is lower. There are myriads of external circumstances—such as gender norms, socioeconomic background, ethnicity, or race—that are associated with unequal opportunities that generate differences in individuals' choices and earnings. Moreover, when people make merit judgements based on earnings or effort information, they often do not know the external circumstances under which earnings and effort choices took place. In this context, the question then arises as to what extent individuals' merit judgements take these unequal opportunities into account.

Andre (2025), Dong, Huang, and Lien (2024), Preuss et al (2022), and Bhattacharya and Mollerstrom (2022) provide persuasive evidence that a majority of individuals neglect the presence of unequal opportunities in their assessment of merit. Even if individuals receive credible information about the fact that disadvantaged workers would perform much better in the absence of unequal opportunities, they largely tend to disregard this information and

base their merit assessment on the *actual performance* of agents in the disadvantaged position. Taken together, this growing literature shows that the disregard of unequal opportunities in merit judgements is robust and arises in different settings and cultures.

4. *The Role of Kindness and Guilt in Social Preferences*

In distributional models of social preferences, the players care only about the final *material* payoff consequences for themselves and others, that is, they are consequentialist models. The players' actions are not affected by how kind or unkind they perceive others' actions or by feelings of guilt and other emotions that may be associated with actions. But players' judgements about others' (un)kindness, their emotions, and intentions are often relevant for players' behaviors and the utility they derive from material payoffs.¹⁵ In view of the potential importance of such judgements, this section presents theories of social preferences that formalize the notions of kindness and guilt—reciprocity theories and guilt aversion theory—and discusses the evidence on the ability of these theories to explain facts beyond what purely distributional models can explain. In addition, we also evaluate the shortcomings of the evidence.

4.1 *Reciprocity*

4.1.1 *Models of Reciprocity*

In a seminal paper, Rabin (1993) applied the tools of psychological game theory (Geanakoplos, Pierce, and Stachetti 1989) to the study of intentions-based reciprocity. The key idea in this research is that people have beliefs about others' beliefs and that these higher-order beliefs can affect their utility. Subsequent research by Dufwenberg and Kirchsteiger (2004) and Falk and Fischbacher (2006) has modified and extended Rabin's model, but the basic framework remains similar. The models assume that the desire to raise or lower others' payoffs depends on the perceived fairness or unfairness of their behavior: the perception of kind intentions leads to kind responses, while the perception of unkind intentions causes unkind responses. In Rabin's model, player i 's utility includes both his material payoff π_i and a nonpecuniary fairness payoff that is the product of player i 's belief about player j 's kindness $\tilde{f}_j$ and player i 's kindness f_i as follows:

$$(11) \quad U_i \equiv \pi_i + \alpha \cdot \tilde{f}_j \cdot f_i.$$

The fairness payoff is given by $\tilde{f}_j \cdot f_i$ and $\alpha \geq 0$ measures the weight given to the fairness payoff. $\tilde{f}_j > 0$ means that i believes that j is kind to her, while $\tilde{f}_j < 0$ indicates that i believes that j is unkind. Likewise, $f_i > 0$ indicates that i is kind to j and $f_i < 0$ means that i is unkind to j . From this it follows that player i can increase his utility by responding to perceived kindness (i.e., $\tilde{f}_j > 0$) with kindness ($f_i > 0$) and to perceived unkindness ($\tilde{f}_j < 0$) with unkindness

¹⁵A small example makes this point: Two brothers are dividing a pie. One cuts it into two pieces and offers the choice of pieces to his brother. The second brother takes the larger piece and the first brother complains that he would have taken the smaller piece. The second brother says: "You got what you wanted. What is the problem?" Perceptions of (un)kindness may well play a role here. The second brother may, for example, infer unkind intentions from the fact that the first brother split the pie unequally.

($f_i < 0$). Note that utility function (11) also implies that j 's payoff enters i 's utility function because the kindness of player i depends on whether (for a given belief of player i about j 's strategy) player i 's strategy gives j a higher payoff than j 's fair payoff.¹⁶

In Rabin's model, mutual cooperation is an equilibrium in the simultaneous prisoner's dilemma because mutual cooperation is associated with mutual kindness, which triggers reciprocal fairness incentives for cooperation. Likewise, mutual defection is an equilibrium involving mutual unkindness. In addition, Rabin's model can explain rejections in the bilateral ultimatum experiment. However, one limitation of his model is that it only considers simultaneous two-player games.

Dufwenberg and Kirchsteiger (2004) develop a theory of reciprocity for n -person extensive form games. Their approach retains Rabin's idea that reciprocal fairness and the associated (un)kindness perceptions provide nonpecuniary incentives for the players. To adapt this idea to extensive form games, they propose a new solution concept—sequential reciprocity equilibrium—and prove an equilibrium existence result. Another advantage of the Dufwenberg and Kirchsteiger model is that it can, in contrast to Rabin's model, explain conditional cooperation of second movers in a *sequentially* played prisoner's dilemma—a behavior that has been widely observed in many experiments. Falk and Fischbacher (2006) combine the notion of inequality aversion with reciprocity concerns, which enables them to explain behavioral phenomena that inequality aversion alone cannot explain.

Finally, Charness and Rabin (2002) also combine distributional preferences with negative reciprocity. In addition, they introduce the notion of “demerit,” which basically means that a player deserves less merit and thus receives less weight in the decision-maker's utility function. A player deserves demerit if he misbehaves relative to a group's normative behavioral standard. Recall that individuals in their purely distributional model care with weight $(1 - \lambda)$ about their own payoff, π_i , with weight $\lambda\delta$ about the payoff of the least well-off player, and with weight $\lambda(1 - \delta)$ about the group's total payoff. In the extended version of their model, the least well-off player's payoff, as well as that of any other player, receives less weight if they misbehave. In addition, a misbehaving player's payoff directly reduces the decision-maker's utility, which causes an incentive to reduce the misbehaving player's income. To capture these motivational forces, Charness and Rabin (2002) introduce three further preference parameters in addition to λ and δ .¹⁷

One objection to reciprocity models and, in particular, to that proposed by Charness and Rabin (2002), is that they are considerably more complex than purely distributional preference models. One might thus wonder whether an increase in complexity is needed. However, it seems fair to say that there are a number of papers that provide data that distributional models cannot explain without considering reciprocity.

Consider, for instance, the following binary ultimatum experiment where the proposer can propose (8 for self, 2 for other) or (5, 5) which is taken from Falk, Fehr, and Fischbacher (2003). In this experiment, the rejection rate of the (8, 2) offer is 44 percent, while in a slightly different experiment, where the equal-payoff alternative (5, 5) is replaced by (8, 2), the first

¹⁶For a given belief of i about j 's strategy, the fair payoff is defined as the average of the lowest and the highest payoff i can give to j .

¹⁷Other interesting models of reciprocity exist, such as those by Levine (1998) and Cox, Friedman, and Gjerstad (2007). We focus, however, on intention-based models of reciprocity because they have led to a considerable empirical literature.

mover clearly has no choice whatsoever and the rejection rate drops to 18 percent. Thus, depending on the alternative to the (8, 2) offer, the responder can make different inferences about the proposer's intention. When (5, 5) is the alternative, a proposer's unfairness intention when offering (8, 2) becomes very visible, while if (8, 2) is the alternative to itself the proposer has no meaningful choice and thus no unkind intention can be inferred. No distributional preference model discussed in Section 3, including fairness ideals models, can explain the rise in the rejection rate from 18 percent to 44 percent when (5, 5) is the alternative, but models such as those discussed above can. We offer several other examples like this below.

At the same time, however, reciprocity models do have drawbacks. They frequently generate multiple equilibria even in very simple games, and some of these equilibria appear un-intuitive. To see this, consider again the binary ultimatum experiment with the (8, 2) and the (5, 5) offer. Here, the strategy pair (8, 2) for the proposer and "accept (5, 5) but reject (8, 2)" for the responder can be a fairness equilibrium (involving mutual hostility) in Rabin's model. However, why should a proposer who knows that the responder will reject (8, 2) with certainty ever make that proposal? Likewise, why should a proposer interpret the rejection of the unfair (8, 2) offer as a hostile act rather than as an understandable response to an unfair offer?

One point is important before we discuss the empirical evidence for reciprocity. Kind or unkind intentions are assumed to play a key role in intention-based models of reciprocity. One would, therefore, expect empirical researchers to have invested a lot of effort in the identification of subjects' kindness perceptions. Unfortunately, however, with a few exceptions (Offerman 2002; Dhaene and Bouckaert 2010), the empirical literature has often ignored this point. Instead of identifying the subjects' (un)kindness perceptions, the experimenters have typically been the judges of what behavior is kind or unkind. Yet, what really matters is what the participants in the experiment consider to be kind or unkind. Thus, if an experimenter assumes, for example, that a particular behavior is kind but the subjects themselves do not perceive it this way, and hence do not reciprocate, one may erroneously conclude that positive reciprocity (i.e., a positive response to a kind action) is absent when in fact the theory predicts the absence of reciprocal responses.

4.1.2 *Laboratory Evidence on Reciprocity*

To identify reciprocity preferences, many lab experiments used bilateral two-stage experiments that are played sequentially. In the main treatment, a human first-mover could make a supposedly kind or unkind choice, while a random device exogenously determined the first-mover's choice in the control treatment so that the human second mover could not infer any intentions.

Charness (1996, 2004) applied this method to a bilateral gift exchange experiment where an "employer" or a random device determines a wage rate w for a "worker," to which the latter can respond with the choice of a costly effort level e that produces a positively valued output for the employer. Charness observes a strong positive relationship between w and e in both treatments. However, the effort level at low wages is lower when the employer chose the wage than when it was generated exogenously, suggesting the presence of negative reciprocity. On the other hand, there was little difference across treatments in the effort levels at high wages. Thus, intentionally receiving a low wage is likely to be seen as unkind because the worker knows that higher wages are possible, and frequently even experiences that higher wages are paid (as the worker faces offers from 10 different employers in sequence). In contrast,

it appears less obvious that higher wages are kind because they may be offered for strategic reasons to elicit a higher effort level. This may help to explain why we observe that positive reciprocation is less prevalent than negative reciprocation.

Offerman (2002) also uses the random versus intentional first-mover choice approach of Blount (1995) and Charness (1996) to study positive and negative reciprocity. He considers players' responses to an unambiguous helpful or hurtful choice. The helpful choice generates a positive payoff for the responder, while the hurtful choice causes a negative payoff. The responders could sacrifice one unit to either increase or decrease the first mover's payoff by four units. In response to an intentional helpful choice, 75 percent of the responders increased the first-mover's payoff while only 50 percent of the responders did so in response to an unintentional helpful choice. The difference of 25 percentage points is not significant but may well reflect the limited number of observations (12). In contrast, the effect of negative intentionality is quite strong: Responders reciprocate 83.3 percent of the intentional versus 16.7 percent of the unintentional hurtful choices. This difference is significant at $p < 0.01$.

To what extent are these large differences in reciprocation patterns a result of subjects' kindness perceptions? Offerman collected proxy measures of responders' (un)kindness experiences by measuring their positive and negative emotions. He finds that intentional hurtful choices generate much stronger negative emotions than unintentional hurtful choices do. In contrast, intentional helpful choices generate about the same positive emotions as unintentional helpful choices do. Moreover, negative (positive) emotions after a hurtful (helpful) first-mover choice are significantly correlated with punishing (rewarding) responses. Taken together, these results suggest that the differences in reciprocation patterns are driven by differences in the extent to which intentional hurtful and helpful choices trigger (un)kindness perceptions. The results are thus nicely in line with reciprocity theory, but also suggest that inducing kindness experiences through intentional choices can be more difficult than inducing unkindness experiences. This differential ease with which one can induce kindness versus unkindness may well have been a factor in the results reported in Charness (1996, 2004).

Several other papers (Brandts and Solà 2001; Brandts and Charness 2003; Falk, Fehr, and Fischbacher 2003; Charness and Levine 2007; Falk, Fehr, and Fischbacher 2008; Dhaene and Bouckaert 2010) support the conclusion that intentions matter for reciprocal responses. The evidence in these papers, as well as those discussed previously in this section, typically suggests that in addition to distributional fairness concerns, intentions are likely to play a role for deviations from self-interested behavior. While several of these papers do not explicitly measure kindness perceptions, they nevertheless implement plausible manipulations of intentions via random versus intentional first-mover choice designs or via other means.

Not all laboratory evidence appears to support the relevance of intention-based reciprocity. Bolton, Brandts, and Ockenfels (1998) find strong evidence for distributional preferences, no evidence for positive reciprocity, and only weak evidence for negative reciprocity. We believe, however, that their findings do not show the absence of reciprocity, but are rather the result of an experimental design that very likely failed to elicit kindness and unkindness perceptions. Participants made choices in 2×6 matrix games via the strategy method. The payoff matrices used for their different treatments are shown in Supplemental Appendix 4. In our view, the presentation of payoffs in this way makes it very difficult to induce any kindness or unkindness judgements for at least two reasons. First, the 2×6 game matrix is so complex that even a researcher might appreciate a guiding hand to understand the (un)kindness interpretations built into the matrix. Second, the game is a simultaneous move game and although the authors

use the strategy method, the kindness judgements only emerge through a complex reasoning chain. This is rather demanding, and presupposes a “theory of mind” capacity that many people are unlikely to have naturally.

4.1.3 *Field Evidence on Reciprocity*

There is evidence from field settings that suggests negative reciprocity is often present. Workers have been known to engage in sabotage, less effort, low-quality work, or increased theft rates after a pay cut or other actions perceived to be unfair (see for example Lawler and O’Gara 1967; Greenberg 1990; Schminki, Cropanzano, and Rupp 2002), particularly when procedural justice in the organization is low (Skarlicki and Folger 1997). The studies by Krueger and Mas (2004) and Mas (2006, 2008) also present results suggesting that workers’ retribute to negative wage changes that are perceived as unfair. The case for positive reciprocity in the field is weaker; cases involving tipping when on the road or higher response rates to mailed surveys that include small gifts may instead reflect guilt aversion, discussed ahead.

Field evidence from uncontrolled natural environments rarely allows researchers to unambiguously pin down motives because reputational and repeated game issues may play a role. Field experiments are, however, capable of controlling these factors. Gneezy and List (2006) conducted such a field experiment that examined the effect of paying more than the participants were led to expect. In experiments in two different locations, they advertised to the public a job with a wage of \$10 (or \$12) per hour, which was also the wage actually paid in the control treatment. However, in the main treatment the people who showed up received, to their surprise, a wage of \$20 per hour. The job was six hours of work, split into morning and afternoon sessions. The research question is whether this surprise overpayment leads to greater effort.

In fact, there was significantly higher productivity in the morning but no effect after lunch. Breaking production into four 90-minute segments, the average production in the gift treatment is 27 percent, 11 percent, 1 percent, and 2 percent higher than in the control treatment, indicating that the treatment effect completely disappears after the first 3 hours. The authors argue that this shows that reciprocity is a weak and ephemeral phenomenon and that paying higher wages may not lead to higher net profits. While we do not dispute the behavioral findings, we note that the authors did not measure the workers’ kindness perceptions. Thus, it remains unclear whether the higher productivity in the morning reflects intention-based reciprocity. Likewise, it remains unclear whether the absence of higher productivity in the afternoon occurred even though workers saw the \$20 wage as kind, that is, whether reciprocity is indeed a very short-lived response.

In this context, a key question is how the recruited subjects interpreted the surprise wage increase. For example, perhaps subjects inferred from the higher wage that the employer is much wealthier than they thought or that the employer made an advertising error when the subjects were initially recruited for \$12. If so, then the initial effort response to the higher wage may not reflect an increase in fairness perceptions, but it may have resulted from short-run effects on subjects’ moods. Bellemare and Shearer (2009) conducted a field experiment on the influence of wage gifts on worker productivity that circumvents this problem by providing a natural explanation for the wage increase. In addition, they increased the statistical power for identifying a wage effect with a within-subject design. Bellemare and Shearer found that the wage gift significantly increased workers’ productivity.

The importance of fairness perceptions in response to a wage increase was documented in Cohn, Fehr, and Goette (2015), who elicited independent measures of fairness perceptions and reciprocity preferences. They show that workers respond to a wage increase with higher effort if two conditions are met: (i) workers perceive the wage increase as an increase in fairness and (ii) workers have reciprocal preferences (that were independently identified in a lab experiment).

At the time Gneezy and List (2006) conducted their studies, \$10 or \$12 was a fair wage for this type of work (prevailing rates were roughly \$8.50), that is, workers were already well-paid at the baseline wage. Perhaps this made it difficult to further increase the kindness perception in the group that received \$20 per hour. This interpretation would be consistent with the “fair wage–effort” hypothesis put forward by Akerlof and Yellen (1990). This hypothesis stipulates that workers respond to wage levels with their effort only if they are underpaid relative to a fair reference wage. Amounts paid above this fair wage are not predicted to have positive effects on effort. This theory is also consistent with results from experiments that tested equity theory (Adams 1963). These experiments often found that overpaying subjects did not increase effort (Walster, Walster, and Berscheid 1977). Thus, the surprise in the Gneezy and List study may be not that the positive treatment effect of a wage gift vanishes over time, but that it was found in the first place.

Kube, Maréchal, and Puppe (2012) consider how the form of payment or reward affects behavior. Their experiment consisted of a three-hour job with an advertised pay of €12. In the baseline treatment, this was done as advertised. In two other treatments, there was a surprise bonus announced; in one case this was €7, and in the other it was a thermos bottle worth €7. There were also treatments where it was clear that there was more effort put into the gift (for example, the additional money was folded into a complex origami).

Kube, Maréchal, and Puppe (2012) find that people are 25 percent more productive in the bottle treatment (and in other treatments where a nonmonetary gift was made) than in the baseline. In contrast, the monetary bonus had only a modest increase of 5 percent over the baseline. There is thus no significant positive reciprocity in response to a higher wage but a great deal of positive reciprocity when the worker is given a gift. Although this paper did not measure subjects’ kindness perceptions, a plausible interpretation of this difference is that giving a gift is much more likely to be perceived as kind. These findings suggest that the “currency of reciprocity” may indeed not be surprise money *that is given without further explanation*.

Esteves-Sorenson (2018) and DellaVigna et al. (2022) also examined the impact of surprise wage increases on workers’ effort in field experiments similar to those in Gneezy and List (2006). Estevez-Sorensen carefully controls for a large number of possible confounds and finds little effect of the wage increase on workers’ output, but—like Gneezy and List—the wage increases implied large overpayments relative to the going market wage and she did not control for workers’ fairness perceptions. DellaVigna et al. estimated workers’ effort cost functions and reciprocity parameters by implementing effort tasks (across experiments) where output was more or less responsive to effort. These changes enabled them to make cleaner inferences on how gifts affect workers’ effort choices. They find that in settings where output is relatively inelastic to effort, such that even sizable shifts in gift-induced prosociality toward the employer would result in small productivity improvements, gifts have little effect on output. In settings where output is more elastic to effort, they find significant effects of the gift on output, but the magnitude of this effect is considerably smaller than what can be achieved with piece rates.

Kube, Maréchal, and Puppe (2013) implemented a wage cut treatment to study the role of negative reciprocity. Since internal review boards might not be happy with a researcher promising a wage and then reneging, they chose their words carefully when they advertised a job. The announcement stated: “The hourly wage is projected to be €15”, leaving some room for later wage changes.¹⁸ There were three treatments: no change in pay (€15), pay reduction (€10), and pay increase (€20). In the pay increase condition, productivity is on average very similar to the no change in pay condition, while there is an immediate and sustained decline in productivity by 21 percent in the pay cut condition, indicating the considerable strength of negative reciprocity.

The limited overall impact of “wage gifts” on output is further corroborated by the field studies of de Ree et al. (2018) and Jayaraman, Ray, and Véricourt (2016). De Ree et al. document that a substantial pay raise for teachers in Indonesia neither increased their students’ test scores nor their own test scores in teacher subject knowledge tests. Jayaraman et al. show that a large pay reform that led to considerable rise in fixed wages among Indian plantation workers resulted in substantial output increases during the first three to four post-reform months, after which output converged to the pre-reform baseline.

This discussion raises the question why there is substantial evidence for gift exchange in laboratory experiments but considerably less evidence from the field. We believe that the different role of inequality aversion in the lab compared to the field is a potentially important reason. Paying a high wage in a typical laboratory experiment implies a strong distributional advantage for the “worker” relative to the “employer.” As shown in the random-wage treatment of Charness (1996 and 2004), higher distributional advantages are associated with higher effort levels, suggesting that aversion to advantageous inequality drives the gift exchange, while positive reciprocity plays little role because reciprocation is not higher in the intentional-wage treatment. In contrast, gifts in the form of higher wages in the field studies are typically not associated with a distributional advantage for the worker relative to a clearly defined individual in the role of a principal because the principal is an organization (e.g., a charity, a firm, a ministry, etc.). This means that advantageous inequality aversion is much less likely to play a role in the field compared to the lab.

This interpretation is further corroborated by the findings in Engelmann and Ortmann (2009) and Bellemare, Kröger, and van Soest (2011). Engelmann and Ortmann conducted a lab experiment where the equilibrium (with self-interested parties) involves a large distributional advantage for the “employer.” Thus, unless the wage gift is huge, the gift still implies a distributional advantage for the employer. As a consequence, they observe little extra effort to wage gifts. Kröger et al. show that reciprocity preferences are weaker than inequality aversion in the general population. They simultaneously estimated structural preference parameters measuring inequality aversion and intention-based reciprocity in a representative sample of the Dutch population that participated in ultimatum experiments with randomly determined versus intentional offers. They find that “inequity aversion tends to be more important than perceived intentions in the population as a whole.” Thus, if the stronger of the two motivational forces (i.e., inequality aversion) is absent in field settings, higher wages may have little effect on effort.

¹⁸The exact German wording was “Ihr Stundenlohn beträgt voraussichtlich €15.”

4.1.4 Summary

What did we learn about reciprocity in this section? First, there are widespread and numerous examples of negative reciprocity in the lab and in the field. On the other hand, positive reciprocity is found in some but not all cases. Why this is the case?

A first important reason could be loss aversion, that is, the notion that losses loom larger than gains. Recall that reciprocity theory defines (un)kindness always relative to a reference point. Kindness is naturally related to receiving more than the reference point, while unkindness means receiving less. The finding of Offerman (2002) that negative emotions after an unkind choice are much stronger than positive emotions are after a kind choice is consistent with this view. A second reason is that certain actions—like surprisingly paying higher wages without providing a plausible explanation for the wage increase—can be interpreted in different ways and may thus not be viewed as kind. This interpretation is in line with the findings in Kube, Maréchal, and Puppe (2012), which indicate that clearly interpretable gifts (payoff-equivalent to a wage increase) trigger strong reciprocal effort responses, while mere wage increases show insignificant effects. When employers want to induce reciprocal effort responses, they must carefully design their gifts, and not just offer more money. We typically do not give money to our spouses at Christmas, but think carefully about the gifts for them.

A third potential reason for the difficulties in inducing positive reciprocity is that a certain minimal level of kindness is taken for granted in many modern societies: People usually hold the door for each other. Typically, if one expects kind or favorable treatment and receives it, there is no strong emotional jolt; on the other hand, if one expects kindness and receives unkind or hurtful treatment, the emotional response is much stronger. This means that the threshold for inducing further kindness perceptions is higher than the threshold for inducing unkindness perceptions.

A fourth potential reason is related to the fact that reciprocity is a cognitively demanding concept because it requires reading other people's intentions. This is much harder than just noticing you received less than another individual. In this context, it is interesting that there appear to be large cultural differences with regard to the extent to which intent is taken into account in moral reasoning. This has been shown by anthropologists who assessed how people judge the badness and punish-worthiness of bad outcomes that either occur accidentally or intentionally (Barrett et al. 2016; Curtin et al. 2020). While intentions played an important role for Western populations, they played a more minor role for many other cultures. These findings suggest that it may be interesting to study the role of intentions versus outcomes in social preferences across different cultures. In addition, except for the Bellemare, Kröger, van Soest (2011) study, very little appears to be known about the distribution of reciprocity in broad population samples.

4.2 *The Role of Guilt Aversion in Social Preferences*

Guilt aversion is based on the idea that people prefer to avoid feeling guilty. The basic idea is that decision-makers experience guilt if they believe they disappointed or “harmed” others whose payoff depends on their actions. Based on this intuition, Dufwenberg (2002) and Battigalli and Dufwenberg (2007) developed a formal theory of guilt

aversion.¹⁹ The notion of guilt due to “disappointing others” or “harming others” in the theory of guilt aversion can be illustrated with the example of a waiter in a restaurant who expects a tip of y for his decent service to a particular customer. Suppose also that the customer believes that the waiter expects a tip of y but only gives him a tip of $x < y$. The theory thus formalizes the customer’s guilt aversion by assuming that she experiences a disutility from guilt of $\gamma(y - x)$, where $\gamma > 0$ is a parameter that captures the customer’s sensitivity to guilt. Thus, guilt is based on the extent to which the customer expects to disappoint or “harm” the waiter relative to the waiter’s expected payoff (i.e., the expected tip). This also means that guilt aversion theory is a theory of social preferences because the waiter’s payoff enters the customer’s utility function through this definition of guilt.

There is a myriad of social interactions in which guilt aversion based on the notion of “disappointing others’ payoff expectations” can play a role. Dufwenberg (2002) first formalized the notion of guilt aversion with the help of an example where a husband can disappoint his wife by divorcing her. But other examples abound. Parents can disappoint their children and vice versa, teachers’ grading can disappoint their students, and business partners, spouses, employers, and employees can disappoint each other; and if the involved parties are guilt averse, the anticipation of these potential disappointments will often change the parties’ behaviors. The avoidance of guilt thus induces prosocial behaviors, in the sense that the potentially disappointed party will receive a higher payoff. However, unlike reciprocity and inequality aversion, guilt aversion cannot easily explain behaviors that reduce others’ payoffs.

While guilt aversion is potentially relevant in many real-world applications, empirically documenting guilt aversion as the underlying mechanism is challenging even in laboratory environments because the decision-makers’ second-order beliefs, which are assumed to drive the decision-makers’ choices, are endogenous, that is, causality is difficult to establish. In addition, the question arises whether it is plausible to assume that any negative deviation from second-order beliefs triggers guilt or whether only negative deviations from “reasonable” beliefs cause guilt. For example, if the waiter expects a tip of 50 percent of a customer’s bill of, say \$100, it appears unreasonable to assume that the customer will experience guilt if she offers a tip of “only” 30 percent of the bill. We will return to this issue toward the end of Section 4.2.

4.2.1 Early Experimental Evidence

The early experimental evidence focused on showing that decision-makers’ prosocial behaviors are positively correlated with their second-order beliefs about the other party’s payoff expectations. Dufwenberg and Gneezy (2000), for example, show this with the help of a lost wallet game. In this game, the finder of a lost wallet can receive a reward, R , from the owner if he returns the lost wallet. The authors measure both the finder’s expected reward, y , and the owner’s beliefs about the finder’s expected reward, $\mathbb{E}[y]$, in an incentivized way. They then show that the actual reward given to the finder is positively correlated with the owner’s

¹⁹The theory has its roots in social psychology: Baumeister, Stillwell, and Heatherton (1994, 1995) advance the notion that people suffer from guilt if they inflict harm on others. Huang and Wu (1994) present the first applied theoretical work in economics to incorporate guilt. The illustrative example in this paragraph describes the notion of *simple guilt aversion*, where one experiences guilt for disappointing another person’s payoff expectations. A second form of guilt aversion is *guilt from blame*, where one experiences guilt only to the extent that one expects to be blamed for a bad outcome (Battigalli and Dufwenberg 2007, 2009). We limit our attention to simple guilt aversion in our analysis.

second-order belief, $\mathbb{E}[y]$. This positive correlation is consistent with the view that guilt aversion drives the owners' reward behavior, but it is also consistent with the existence of a social norm of rewarding finders that affects both the owner's behavior and the parties' beliefs about rewarding an honest finder.

One of the attractive features of guilt aversion theory is that it can rationalize the large positive impact of communication opportunities on cooperation behavior in social dilemmas because the subjects typically make verbal promises to cooperate before the game is played but these promises are subsequently not enforceable (Sally 1995). Nevertheless, communication treatments typically generate much higher cooperation rates compared to a no-communication treatment, which is consistent with guilt aversion if the promising player assumes that she will disappoint the opponent when defecting after a promise.

Charness and Dufwenberg (2006) show this by studying the game where player 1 can enter a trade that involves the sequential exchange of goods (see Figure 3). If player 1 does not trust player 2 and so forgoes the trading opportunity, each party receives a payoff of 5. In case player 1 trusts by sending his goods to 2, player 2 can send her goods to player 1 ("share") or keep her goods ("grab"). If player 2 keeps her goods, player 1 receives nothing, while player 2 has a payoff of 14. If player 2 sends her goods and the batch arrives, which happens with probability $5/6$, the payoffs are (12, 10). However, the sent goods may not arrive with probability $1/6$, in which case player 1 receives nothing but player 2 still receives 10. Note that in this game player 1 cannot distinguish whether bad luck or bad behavior by player 2 generated a payoff of zero.

In this game a self-interested player 1 who expects to face a self-interested player 2 will never trust. If player 2 is guilt averse, however, then it may be rational for player 1 to trust. From the viewpoint of guilt aversion theory, the key question is whether player 2's belief about player 1's expectation about 2 choosing *share* is positively correlated with player 2's actual probability of choosing *share*. Guilt aversion theory predicts that higher beliefs by player 2 about 1's expectation will increase the probability that player 2 will actually choose *share*.

Charness and Dufwenberg gave player 2 the option to send free-form messages to player 1 because they conjectured that this may be a particularly powerful way of strengthening player 2's second-order belief. The reason is that player 2 has a strong incentive to convince player 1 that 2 will share by making a promise. Intuitively, if player 2 tries to persuade player 1 that he will choose to share, then player 2 can hardly avoid the inference that in case of trust player 1 had a high belief that 2 will share. Thus, guilt aversion also provides a mechanism that can explain why communication between parties often enhances cooperative behavior.

The experimental results clearly show the effectiveness of communication. As expected, players 2 often, but not always, promised to share and these promises were very effective: Players 1 chose *trust* 79 percent of the time and 2s chose *share* 79 percent of the time when a promise was made. This compares to 37 percent and 33 percent, respectively, when no promise was made. In addition, the authors observe a strong relationship between 2's belief about 1's belief that 2 will choose *share* and 2's actual choice of whether to *share*. These beliefs were significantly higher for 2s who chose *share* compared to 2s who chose *grab* (66.7 percent versus 42.7 percent)—which is consistent with guilt aversion theory.

Vanberg (2008) argued, however, that player 2 may keep her promise not because of guilt aversion but because she simply experiences an obligation to keep her promise. Vanberg developed an experimental design that distinguishes between the guilt aversion and the obligation hypothesis. His evidence suggests that feeling obliged to keep promises is indeed a driver of player 2's behavior.

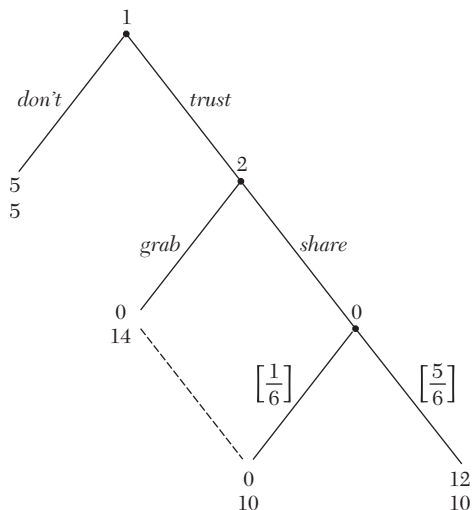


Figure 3. Sequential Exchange with Moral Hazard

4.2.2 Overcoming Problems in Identifying Guilt Aversion

A key issue in identifying guilt aversion by measuring players' second-order beliefs is that these beliefs are endogenous. In the sequential exchange game in Figure 3, the empirically measured beliefs of player 2 about 1's expectation could easily be a result of 2's choice instead of driving it. For example, if 2 believes that player 1 correctly anticipates her choice, then 2's second-order belief coincides perfectly with 2's action. In addition, because players' second-order beliefs are typically elicited after the players have made their choices, the beliefs may simply reflect an ex post rationalization of the choices made.

Several approaches have been put forward to solve this identification problem, such as the disclosure approach and approaches that experimentally manipulate second-order beliefs. Applied to a dictator experiment, for example, the disclosure approach elicits the recipient's expectations about the dictator's transfer and informs the dictator about this expectation. Thus, there is no need to elicit the dictator's second-order belief because the experimenter knows that belief directly. The disclosure approach has been implemented in several papers and led to mixed evidence for guilt aversion. Some papers found no evidence for a positive correlation between second-order beliefs and the players' prosocial choices (Ellingsen et al. 2010; Bellemare, Sebald, and Suetens 2017), while others found a significant correlation (Reuben, Sapienza, and Zingales 2009; Bellemare, Sebald, and Strobel 2011).

The disclosure approach is not without drawbacks, however. While it circumvents the endogeneity problem associated with the direct elicitation of second-order beliefs, it introduces the possibility of other confounds. For example, providing information about a partner's expectation may also signal something about the prevailing social norm, implying that guilt aversion may be confounded with social norm compliance. In addition, if players are informed about the other players' expectations, they may wonder why the experimenter does

this, whether the other players know this, and if so, whether other players have biased their expectations strategically.

Is there a way to overcome the difficulties in reliably identifying guilt aversion through the exogenous variation of second-order beliefs? The papers by Ederer and Stremitzer (2017) and Khalmetski (2016) indeed solve this problem. Due to space constraints, we focus on the first paper. Ederer and Stremitzer use a variation of the sequential exchange game displayed in Figure 3. As in Charness and Dufwenberg (2006), player 2 can send free-form messages to player 1 before player 1 decides whether to trust.

Ederer and Stremitzer successfully implement an exogenous variation in beliefs by introducing a chance move that generates either a reliable or an unreliable trading technology in case that player 1 plays *trust*. If the trading technology is reliable, there is a 5/6 probability that player 2 has the option to trade (i.e., the option to play *share*), while in case of an unreliable technology the option to choose *share* only materializes with probability (1/6).²⁰ Importantly, once the type of trading technology is determined, both players are informed about it, which will obviously affect the players' expectations. In particular, player 1 will expect a higher payoff in case of a reliable technology than in an unreliable trading technology because player 2 has the option of choosing *share* with a high probability in the former case. Moreover, player 2 will know this and will thus have higher second-order expectations which, according to guilt aversion theory, should trigger a higher rate of share choices in the presence of a reliable technology.

Ederer and Stremitzer (2017) verify that the reliable trading technology indeed induces higher second-order beliefs regarding the probability of playing *share*. Moreover, this exogenous increase in second-order beliefs holds regardless of whether player 2 made a promise, sent no message, or merely engaged in empty talk. Guilt aversion thus predicts that this exogenous increase in beliefs will be associated with a higher percentage of *share* choices under a reliable technology regardless of the message player 2 sends. Interestingly, however, the percentage of sharing choices is only higher under the reliable technology *in case of a promise*. When player 2 made no promise or simply engaged in empty talk, the exogenous shift in second-order expectations was not accompanied with an increase in choosing *share*.

These results may help us understand the conditions under which guilt aversion theory applies. It seems that moving second-order expectations is not always enough to move behavior in the direction of guilt aversion, but that additional psychological conditions—such as a promise—need to be present to induce guilt averse behavior. The special status of promises is also supported by evidence in Bracht and Regner (2013), who find a significant effect of promises even after controlling for second-order beliefs.

The theory of guilt aversion is devoid of any explicit notion of fairness or justice. The only thing that enters the guilt calculus is whether the decision-maker falls short of the other player's payoff expectations *regardless of whether these expectations are normatively justified or not*. In a dictator game, for example, the dictator's guilt will be determined by how much the actual transfer to the recipient falls short of the dictator's second-order belief about the recipient's expected transfer. However, it is known that in anonymously played dictator games where players are randomly assigned to their roles, many subjects believe that the available money should be shared equally (Fehr and Fischbacher 2004). Thus, one would expect the dictator to

²⁰Ederer and Stremitzer also allow for different degrees of sharing, but we abstract from this to keep the exposition simple.

feel guilt only if she gives less than 50 percent. According to the purely belief-based approach to guilt aversion, however, guilt also arises if a player would have an unreasonably high expectation of receiving a transfer of, say, 70 percent. But is this approach really reasonable?

The paper by Andreoni and Rao (2011) provides interesting insights into this question. The authors conducted dictator games where the recipients could stipulate a numerical amount that they would like to receive. A key finding was that recipients who asked for more typically received greater transfers, but this only held within limits because subjects who demanded more than the equal division were punished with considerably lower transfers. This finding is consistent with the view that beliefs about other players' expectations will only enter the guilt calculus if the expectations are "reasonable" or in line with prevailing social norms. When viewed from this perspective, the theory of guilt aversion can be interpreted as a theory of social norm enforcement where the norm itself is taken as exogenous. In this view, social norms govern people's expectations and guilt only arises to the extent to which a player does not meet the prosocial behaviors that the norm determines.²¹

4.2.3 *Summary*

The theory of guilt aversion provides a simple and intuitively powerful account of prosocial behavior. There is considerable behavioral evidence that is consistent with guilt aversion, and behavioral measures of individuals' guilt aversion have been shown to be correlated with psychological measures of guilt proneness (Bellemare, Sebald, and Suetens 2019). However, the rigorous identification of behavioral guilt aversion is nontrivial and involves difficult identification problems. Guilt aversion also does not appear to manifest itself in all environments. Guilt aversion most likely plays a role in environments where social norms guide expectations and what is considered appropriate behavior. Guilt aversion is, therefore, best viewed as a psychological mechanism of norm enforcement.

5. *The Role of Self-Image and Social Image Concerns*

One's image can be important for a variety of reasons. Just like looking in a mirror and adjusting physical appearance, people will take actions to appear more favorably. This applies to both self-image and social image, which to some extent seem inextricably intertwined: improving one's self-image or self-esteem may well have positive spillovers socially, and one's social image can lead to a better self-image. This topic was first considered in the field of psychology, where impression management, self-concept, and self-presentation are terms that reflect the care taken to appear more favorably to self and others. The Tesser (1988) self-evaluation maintenance model proposes that people are motivated to maintain both positive self-views as well as their perception about how other individuals view them. Overall, there is considerable evidence indicating that social comparisons affect self-esteem (Gastorf and Suls 1978; Molleman, Pruyn, and van Knippenberg 1986).

Self-perception theory (Bem 1972) has provided an intellectual basis for the modeling of self and social image concerns; it stipulates that people observe their own behavior to infer what they are thinking and how they are feeling. People's behavior may therefore also serve the purpose

²¹ On this issue see also Cartwright (2019) who provides an interesting discussion of the evidence for belief-based guilt compared norm-based guilt.

of self-signaling and social signaling. One way of formalizing self-signaling is to assume that it is an attempt to influence the beliefs of a future self who cannot remember the original motivation for the behavior. Prelec and Bodner (2003) were the first to develop a dual-self-signaling model, an approach that was later also applied in Bénabou and Tirole (2006, 2011), Grossman (2015), and Grossman and van der Weele (2017).²²

Consider, for example, a simplified version of Bénabou and Tirole's (2006) model. Here, an agent has an intrinsic valuation, v_a , for contributing a to a charity that generates a cost, $C(a)$, so the direct benefit of choosing a is $v_a a - C(a)$. Because the agent cares about her social image, there are also reputational costs and benefits, R , that depend on what others infer on average about the agent's intrinsic prosociality, $\mathbb{E}[v_a | a]$, given the agent's observable prosociality a :

$$R(a) \equiv x \gamma_a \mathbb{E}[v_a | a], \quad \text{with} \quad \gamma_a \geq 0.$$

The sign of γ_a indicates that people would like to appear prosocial and $x > 0$ is the visibility of their actions. Note that $R(a)$ could represent an instrumental or an affective value of one's social image. Letting $\mu_a \equiv x \gamma_a$, an agent with reputational concerns chooses a to satisfy:

$$(12) \quad \max_{a \in A} \{v_a a - C(a) + \mu_a \mathbb{E}[v_a | a]\}.$$

In a nutshell, this model assumes that individuals care for the intrinsic value of prosociality, $v_a a$, the material cost involved in behaving prosocially, $C(a)$, and their social image, $\mu_a \mathbb{E}[v_a | a]$. They choose a to optimally balance these components. The model can also be applied to self-image concerns by assuming that there is some chance that one might later fail to remember or (remember in a self-serving manner) the reasons for making a choice. If observable actions are easier to remember than imagined motives, signaling our type to our later self with our current actions could make sense. Perhaps the exact feelings or signals at that time are obscured with some probability proportional to x and the agent later cares about "how prosocial she is".

In the following, we discuss how concerns about one's image—whether self or social—affects *desirable* behaviors. Because *social* desirability is intrinsically tied to the prevailing social norms, a concern for one's social image generates social pressure to conform to these norms. Social image models are therefore essentially models of social norm compliance. In contrast, self-image concerns are based on an individual's *private* identity, the kind of person an individual wants to be, which could differ from the prevailing norms. However, in practice social and private desirability will often coincide.

Note also that social image concerns cannot easily explain the prosocial and the punishing behaviors observed in paradigmatic economic experiments (see Box 2) because social pressure could play little role in the anonymous environments implemented in these settings.²³

²²Battigalli and Dufwenberg (2022) make a strong case that image concerns imply that the decision-maker's belief about other's beliefs enters the utility function. In other words, proper modelling of image concerns requires the tools of psychological game theory. In the case of self-image concerns, beliefs about other self's beliefs, and in case of social image concerns, beliefs about other people's beliefs enter the utility function. To ease exposition, we use a simplified version of the Bénabou–Tirole (2006) model.

²³These experiments implemented anonymous interactions among the subjects. Moreover, the lack of subject–experimenter anonymity has only minor, insignificant effects in dictator, ultimatum, and trust experiments (Bolton, Katok, and Zwick 1998; Barmettler, Fehr, and Zehnder 2011).

However, could it be that self-image concerns can explain some of the prosocial behavior in these experiments?

5.1 *The Evidence for Self-Image Concerns*

Dana, Weber, and Kuang (2007, henceforth DWK) conducted a paradigmatic and influential study providing evidence that appears to be consistent with self-image concerns in altruistic choice. They implemented a binary dictator experiment in a full information treatment and a hidden information treatment. In the *full information treatment*, the allocator can choose between option A, yielding allocation (6, 1), and option B, with allocation (5, 5). In the *hidden information treatment*, the allocator has again two options (A and B) with corresponding payoffs of 6 or 5 for herself but unknown payoffs for the other party. There are two possible states of the world. In one state, the players' preferences over A and B are aligned because choosing A yields payoffs of (6, 5) while choosing B leads to (5, 1). There is a conflict of interest in the other state because the payoffs are the same as in the full information treatment. In the hidden information treatment, the chooser does not know which state of the world prevails but she has the option to reveal the true state at no cost by clicking a box on the computer screen.

In the full information treatment, 74 percent of the dictators chose the fair (5, 5) option, while only 37.5 percent of the dictators who were randomly assigned to the conflict state chose the fair option in the hidden information treatment. Thus, despite the fact that dictators in the conflict state of the hidden information treatment faced identical cost and benefits from a fair choice compared to the full information treatment, the moral wiggle room that is present in the hidden information treatment seems to have induced them to behave much less fairly. A reason for this is that only 50 percent of the dictators who were assigned to the conflict state actively chose to seek information about the true state. All of those who remained uninformed chose the selfish option, while the informed dictators chose the fair option in 75 percent of the cases.

To what extent can self-image concerns explain willful ignorance? A key paper in this regard is Grossman and van der Weele (2017), who developed a multiple-self model where a decision-maker (DM) self manages her image vis-à-vis an observer self. One can illustrate the basic intuition of their paper with the help of the notation used when describing the simplified Bénabou and Tirole (2006) model above where the decision-maker's utility is given by

$$v_a a - C(a) + \mu_a \mathbb{E}[v_a | a, \sigma].$$

The *DM self* knows her preferences and derives altruistic utility, $v_a a$, from behaving prosocially ($a = 1$), and also puts a positive weight, μ_a , on her self-image, $\mathbb{E}[v_a | a, \sigma]$. The *observer self* lacks introspective knowledge of the DM's altruism parameter, v_a , but can infer it from the DM's actions and knowledge about the state of the world. The state of the world, s , is either unknown, the conflict state, or the aligned state. The observer's inference about the DM's altruism is given by $\mathbb{E}[v_a | a, \sigma]$, which is the basis for the DM's self-image.

The key insight derived from this model is based on the existence of a "willful ignorance equilibrium" in which the selfish individuals ($v_a = 0$) behave prosocially in neither the full information treatment nor in the hidden information treatment ($a = 0$). Altruistic individuals, however, can be sorted into two groups: those with a strong preference for altruism ($v_a \geq$

$v^* > 0$) and those with a weak preference for altruism ($v_a < v^*$). The strong altruists (i) choose the prosocial option in the full information treatment and (ii) acquire information and choose the prosocial option in the conflict state of the hidden information treatment.

The third category of players, the weak altruists, are necessary to explain willful ignorance in the DWK paper. In the full information treatment, the DM and the observer know that the conflict state prevails. Choosing option A with payoffs (6, 1) thus provides a clear signal to the observer that the DM's altruism is low, which hurts the DM's self-image. In the hidden information treatment, however, neither the DM nor the observer know the state of the world. Therefore, maintaining the uncertainty and choosing A has a much lower signaling value regarding the DM's altruism because there is a 50 percent probability that choosing A was not an unfair choice. But the *critical assumption* here is that the *observer does not infer something negative* about the DM's altruism from the mere fact that *the DM decides to remain ignorant*. In this regard, self-image models are also self-deception models because individuals can fool themselves into believing that they are more prosocial than they in fact are. This kind of self-deception has been observed, for example, in Carlson et al. (2020), who report that a non-negligible share of individuals recalls being more generous in the past than they actually were—an effect that occurred mainly in those individuals who violated their own fairness standards.

Thus, the essence of the model is that the hidden information treatment enables the DM to reduce the negative signaling value of behaving unfairly by obfuscating the observer's inferences about her altruism in case of an unknown state of the world. For weak altruists, the total value of an altruistic act—which consists of intrinsic altruistic utility plus the signaling value—thus declines below the cost of the altruistic act.

The previous considerations indicate that self-image concerns are consistent with the evidence in DWK but they do *not* show that individuals remain ignorant *because* of self-image concerns. In fact, a recent article by Exley and Kessler (2023) documents that self-image concerns can at most explain a minority of willful ignorance choices. These authors conducted a series of six different studies with roughly 6,400 individuals. They introduced a new condition that is identical to the hidden information condition in DWK except that the allocator is now an impartial third party who allocates payoffs between *two other* parties.

The authors compare this other/other condition with the traditional self/other condition where the allocators' self-interest is at stake. Note that in the other/other condition self-image concerns about the allocator's selfishness cannot affect choices because the allocator's self-interest is not at stake. Therefore, if information avoidance in the self/other condition is driven by the desire to reduce signals of selfishness, information avoidance should completely vanish in the other/other condition. However, in contrast to this prediction, depending on the study, only between 14 percent and 34 percent of information avoidance vanishes in the other/other condition. In addition, Exley and Kessler (2023) show with additional treatments that, rather than being driven by self-image concerns, information avoidance is largely driven by subjects' aversion to making a trade-off between two participants, their desire to avoid the bad news of being in a nonaligned state, and their inattention, confusion, or laziness.

The limits of self-image driven information avoidance under hidden information is also indicated by Grossman (2014), who examined the robustness of willful ignorance to small, seemingly innocuous changes in the experimental design. He changes the default for learning

the true state in the hidden information treatment. In their “default non-revealed” treatment, which is similar to DWK, subjects need to click a box to receive information; they remain uninformed otherwise. In “active choice” treatment, participants must click on a box to either learn the state or to not learn the state. In the “default revealed” treatment, the default is reversed so that participants learn the state unless they click on a box to say they don’t want the information.

The data in Grossman (2014) confirm the basic result in DWK: About 45 percent of the participants in default non-revealed maintained their ignorance and then almost invariably chose allocation (6, 1). Yet matters change dramatically when one must make an active choice of one of the two boxes, with the non-reveal rate dropping significantly to 25 percent. The most compelling evidence, though, comes from reversing the default. When one is required to affirmatively state that one does not wish to learn the true state (as in default revealed), only 3 percent of the participants opt not to learn it. This is a remarkable departure from the rate in default non-revealed, and it suggests that self-image-driven willful ignorance is not a very robust phenomenon.

Finally, to what extent is the reduction in altruism due to hidden information a robust phenomenon? A recent meta-analysis of a large number of studies similar to those of DWK (Vu et al. 2023), comprising roughly 6,500 individuals and 56 treatment effects, shows that the role of hidden information for altruistic choice is considerably smaller than in DWK: In DWK 74 percent of individuals chose altruistically in the full information treatment but only 37.5 percent did so in the conflict state of the hidden information treatment. This large decline in altruistic behavior under hidden information is due to the 50 percent of individuals who remain uninformed *and* behave selfishly. In contrast, in the meta-analysis 57 percent make the altruistic choice under full information while under hidden information this drops to 41.4 percent, indicating a much smaller reduction in altruism. It is, however, important to keep in mind that, given the findings in Exley and Kessler (2023), only a relatively small part of this smaller decline in altruism can be attributed to self-image concerns.

Carpenter and Robbett (2022) conducted another robustness test examining how moral wiggle room affect the estimates of structural distributional preference parameters. In their experiment, the subjects participated in 45 different binary dictator experiments that included many different costs of altruistic choices. In addition, subjects also faced many situations where they could reduce the other’s payoff at a cost to themselves. This set-up is likely to decrease demand effects associated with standard single shot dictator experiments because the large variation across choice situations provides many justifications for selfish behaviors (e.g., high costs of altruism in some situations). Carpenter and Robbett show that the existence of moral wiggle room does not lead to a significant change in the estimated distributional preference parameters. This finding is important because it indicates that moral wiggle room may play little role in the typical experiments (Fisman, Kariv, and Markovits 2007; Fisman et al. 2015; Fehr@Epper@Senn 2023a) used to estimate distributional preference parameters.

The literature on information avoidance in moral wiggle room experiments has undoubtedly increased our knowledge about the intricacies and driving forces underlying prosocial behavior. The overall evidence indicates, however, that the influence of self-image concerns is limited and that phenomena such as willful ignorance, initially attributed to self-image concerns, are not very robust; they can be overcome with small changes in choice architecture, such as requiring active choice or changing defaults.

5.2 The Evidence for Social Image Concerns

The formal framework in Bénabou and Tirole (2006) can be applied to self-image concerns and to social image concerns. In the latter case, the multiple-self interpretation can be discarded and the term $\mu_a \mathbb{E}[v_a|a, \sigma]$ represents the decision-makers' intrinsic valuation of being viewed as a prosocial actor by relevant third parties. As we will see below, there is strong evidence for the impact of social image concerns on prosocial behaviors, and many authors have provided models of such concerns (Holländer 1990; Bénabou and Tirole 2006; Ellingsen and Johannesson 2007, 2008).

Social image concerns are, in a sense, a preference for social approval. Rege and Telle (2004), for instance, conducted a simultaneously played one-shot public goods experiment where complete free riding is the dominant strategy. They implemented a private condition and a public condition where all group members saw how much an individual had contributed. In both conditions each individual first privately committed to a contribution level by deciding how much of her monetary endowment to put into an envelope dedicated to the public good. Thereafter, each subject had to stand up and put her envelope into a box in front of a room. In the private condition the envelopes remained sealed; In contrast each subject had to open her envelope, count the money in the envelope, and write the amount contributed on a blackboard in the public condition. Obviously, the public condition generates strong social image/approval incentives such that the average contribution rate increased from 34 percent in the private condition to 68 percent in the public condition.

Ariely, Bracha, and Meier (2009) also exploit the distinction between a private setting (where choices remained private), and a public setting (where choices were made public to the other participants in the study) to examine social image effects on prosocial behavior. In their lab study participants performed a real-effort task (pressing two keys sequentially) that led to donations on their behalf to the American Red Cross. In the public setting, the subjects worked much harder to generate donations to the Red Cross. In addition, the authors tested an interesting prediction that follows from several social signaling models: that monetary incentives crowd out prosocial behaviors driven by social image concerns because public observers will attribute at least a part of subjects' effort to the monetary incentive rather than to their prosociality. Ariely, Bracha, and Meier confirm this prediction neatly.

Andreoni and Bernheim (2009) consider social signaling in a \$20 dictator experiment with a stochastic element to the decision. Specifically, there is some probability p that nature transfers a fixed value, x_0 , to the recipient, that is, nature overrides the dictator's decision. To heighten social image concerns, participants were informed that all participants and outcomes would be publicly identified at the end of the session. Note that only the *outcomes* for the two players but *not* the dictators' *choices* are made public. This means that selfish dictators can hide their choices in those dictator experiments in which nature chooses a low transfer with positive probability, and thus avoid damaging their social image by making the same transfer that nature would have made.

Andreoni and Bernheim elicit dictators' choices for four values of p (0, 0.25, 0.50, and 0.75) and two values (0 and 1) for nature's fixed transfer. Their findings provide strong support for the idea that people care about their social image. Between 57 percent (in case of $x_0 = 0$) and 69 percent (in case of $x_0 = 1$) of the dictators chose the equal split when $p = 0$. However, the frequency of equal splits dramatically declines with increasing p , and more and more dictators hide behind nature's fixed value and choose exactly x_0 . For example, for $p = 0.5$

and $x_0 = 0$, 72 percent of the dictators choose exactly 0 and only 28 percent the equal split. Likewise, for $p = 0.5$, and $x_0 = 1$, roughly 45 percent of the dictators choose $x_0 = 1$ and the frequency of equal split is slightly below 40 percent. Thus, taken together, the evidence from laboratory experiments suggests that social image concerns (being viewed as fair) constitute a strong motive.

In recent years, there has also been increasing evidence from the field suggesting that social image concerns can have powerful effects on important economic behaviors such as labor supply (Bursztyn, González, and Yanagizawa-Drott 2020), education investments (Bursztyn and Jensen 2015), conspicuous consumption (Bursztyn et al. 2018), effort in the workplace (Mas and Moretti 2009), charitable donations (DellaVigna, List, and Malmendier 2012),²⁴ or voting (DellaVigna et al. 2017). In the latter paper, the authors left flyers on the doors of homes informing the households that they would return the next day to conduct a survey. Half of the households were randomly informed that the survey would ask them whether they voted in a recent election, while the other half did not receive this information. Under the plausible assumption that people who did not vote want to avoid both lying to an interviewer *and* admitting that they did not vote, one would expect that among the informed households the willingness to open the door the next day would be lower compared to the uninformed households. The reason is that by not opening the door individuals can avoid both lying *and* the stigma or shame of being seen as a low civic type. Consistent with this hypothesis, the informed households were indeed 20 percentage points less likely to open the door.

5.3 Summary

The evidence suggests that social image concerns can have strong effects on behaviors that are clearly normatively desirable, while the evidence for self-image driven prosocial behaviors is relatively limited. In contrast to other-regarding preferences like altruism, inequity aversion, reciprocity, or guilt aversion, self and social image concerns share a more self-regarding flavor. This does not mean, however, that truly other-regarding preferences and image concerns are mutually exclusive. In fact, it appears quite plausible that individuals simultaneously hold various other-regarding and image-based motives. In terms of quantitative importance, a recent meta-analysis finds that other-regarding motives appear to be considerably more important than self-image concerns (Vu et al. 2023). Moreover, self and social image concerns can hardly explain behaviors such as (i) rejections of low offers in ultimatum experiments, (ii) the payoff-reducing behaviors observed when subjects face positively sloped budget lines, or (iii) the punishing behaviors observed in many social dilemma experiments, because these behaviors lack a clear prosocial meaning. This view is supported by evidence in Rockenbach and Milinsky (2011), who gave subjects the opportunity to reveal their contributions or their punishments in an anonymous public goods game. Note that the subjects still maintained their personal anonymity when they revealed these behaviors because subjects only knew each

²⁴ In the field experiments of DellaVigna, List, and Malmendier (2012), flyers that told individuals that somebody will come to solicit donations for a charity at a particular time were posted on the front doors of some households. In the control treatment, no flyers were distributed. Flyers significantly reduced the share of households that opened the door and they also reduced overall donations, indicating that some people dislike the social pressure associated with door-to-door funding campaigns. This does, however, not mean that the households that refused to open the door are non-altruistic or generally unwilling to give to charities. It only means that they dislike social pressure to give to *unknown* charities or the *specific* charities for which the study collected money.

others' identification numbers. Interestingly, while individuals were keen to make high contributions publicly visible to the other players in the game, they avoided making their explicit punishments and their low contributions publicly visible. In addition, Balafoutas, Nikiforakis, and Rockenbach (2014) show that subjects who have the opportunity to reward those who punished norm violators refrain from doing, so while it is well-known that many subjects reward those who helped others (Engelmann and Fischbacher 2009).

Image concerns imply that situational and institutional factors that are completely irrelevant and innocuous in a world without image concerns can become highly relevant. Self-image concerns can be mobilized for prosocial behaviors by removing uncertainties about the prosocial effects of these behaviors, by reducing opportunities that enable moral wiggle room, or by making subjects' decisions to remain willfully ignorant known to the other players. All these precautions reduce the likelihood that players can deceive themselves into believing they are more altruistic than they in fact are. Likewise, making decisions transparent and publicly known to relevant others activates social image concerns.

6. *Economic and Political Consequences of Social Preferences*

Like time and risk preferences, social preferences have broad implications for a wide variety of domains. In this section, we review evidence on how social preferences affect (i) cooperation and (ii) employees' responses to wage inequality and wage cuts, and (iii) the extent to which this affects firms' employment decisions. In addition we discuss (iv) the allocation of workers with varying prosociality levels to different industries, (v) how social preferences influence incentives and contracts, and (vi) how they affect participating in protest movements against dictatorships and the political demand for redistribution. These topics do not exhaust the consequences of social preferences, but space constraints force us to limit consideration on the aforementioned issues.²⁵

6.1 *The Role of Social Preferences in Cooperation*

There is a large theoretical and empirical literature on the role of social preferences in cooperation and collusion, including several review papers (e.g., Van Lange et al. 2014; Fehr and Schurtenberger 2018; Balliet, Mulder, and Van Lange 2011). Theory suggests that altruistic preferences tend to facilitate cooperation, while the role of disadvantageous inequality aversion or negative reciprocity is more nuanced and depends on the possibility and the specific features of peer punishment opportunities. If public goods experiments offer these opportunities, disadvantageous inequality aversion and negative reciprocity are motivational forces that facilitate the punishment of free-riders, which helps maintain cooperation levels (Fehr and Schmidt 1999; Fehr and Gächter 2000; Falk and Fischbacher 2006). However, if these opportunities are absent, these forces may have detrimental effects on cooperation and induce players to cease cooperating.

Recent field studies on large-scale cooperation have also established the relevance of peer pressure and peer punishment. Breza, Kaur, and Krishnaswamy (2019) studied whether large groups of decentralized workers cooperate to prevent downward pressure on wages. They

²⁵ There is, for example, a considerable literature documenting the importance of social preferences for individuals' willingness to invest in a socially responsible manner (Riedl and Smeets 2017; Bauer, Ruof, and Smeets 2021).

implemented a field experiment in 183 local labor markets in rural India and show that almost none of the agricultural workers are willing to accept jobs below the prevailing wage when other workers can observe this choice. In addition, they document that this unwillingness to accept low wages is due to workers' willingness to sanction those who accept wage cuts. However, if acceptance of low wages is not observable (and thus not subject to peer sanctioning) the willingness to undercut the going wage increases substantially. Moreover, consistent with the aggregate implications of downward rigidity, Breza et al. also show that measures of social cohesion in local markets correlate with downward wage rigidity and its employment effects across India.

The hugely successful recruitment of soldiers for the British Army at the beginning of World War I, when the army relied entirely on the voluntary recruitment of soldiers, provides another powerful real-world example of large-scale cooperation supported by peer punishment. Roughly 479,000 volunteers were recruited between August 1914 (when Britain declared war on Germany) and September 1914, and approximately 2.5 million men had voluntarily joined the British Army by December 1915. Those who did not join faced the contempt of their community members, who attached big red patches to the free riders' front doors at night, so that everybody could see that the person living there was a dodger (Simkins 1988). Recently, Becker (2022) documented how young women publicly shamed young men who refused to join the army. In many towns and cities, the women handed out white feathers to men in civilian clothes, marking them out as cowards. The young women often took substantial risk when doing so because the affected men retaliated. Becker collects evidence from local newspaper articles and exploits the gradual spread of the movement to show that during the 10 days after the first mention of White Feather Girls in the news, volunteering surged by one-third.

However, the fairness concerns inherently embodied in social preferences can also weaken young men's willingness to voluntarily join the army (Jiang 2024). In a fascinating paper, Jiang (2024) shows—on the basis of individual-level administrative records and by using the draft lottery as a source of exogenous variation—that draft dodging by the rich undermined voluntary enlistment of poorer neighbors in the United States during World War II.

Cooperation is not always a good for the overall society. The cooperation between companies for the purpose of maintaining high prices and the cooperation within criminal organizations are examples. Another example is vote buying, a frequent practice in many countries with weak democratic institutions. In a fascinating study, Finan and Schechter (2012) document how social preferences for reciprocity facilitate vote buying in municipal elections in Paraguay. Vote buying constitutes a serious puzzle in a secret voting environment with selfish voters because they cannot ultimately be forced to vote for the candidate who tries to buy their vote. A selfish voter would just take the bribe and merely claim that he or she voted for the bribing politician. However, this commitment problem can be overcome if voters have an intrinsic preference for reciprocity.

In Paraguay, politicians hire respected community leaders in each village to interact with voters and offer them money and other forms of aid for the promise of their vote. Finan and Schechter (2012) show that these community leaders have a very good knowledge of individual voters' preferences for reciprocity and preferentially target reciprocal voters for vote buying. A one standard deviation increase in reciprocity (measured by the change of trustees' back-transfers in a trust experiment in response to changes in trustors' investments) increases the likelihood of being targeted for vote buying by 44 percent—a finding that is robust to a

large set of controls, including other social preferences and voters' network relationships in the village.

Another domain where social preferences matter is the role of individual heterogeneity in public goods provision. The theory of inequality aversion predicts that groups that are more heterogeneous—in terms of their wealth (endowments) or in terms of the benefits they derive from public goods—are less likely to achieve and maintain successful cooperation, a prediction that a large experimental literature supports (Chan et al. 1996; Chan et al. 1999; Anderson, Mellor, and Milyo 2008) and that is consistent with field observations (Mayer 2001; Fajnzylber, Lederman, and Loayza 2002).

The causal influence of prosocial preferences on groups' abilities to maintain high levels of cooperation has been documented in Gächter and Thöni (2005). They measured individuals' willingness to cooperate in a one-shot social dilemma experiment and subsequently formed three types of homogeneous groups: (i) groups comprising individuals with a high prosocial preference, (ii) intermediate groups, and (iii) groups comprising selfish individuals. When subjects played the one-shot social dilemma, they did not know that they would subsequently be allocated to these groups. Gächter and Thöni show that aggregate group cooperation in public good experiments with a dominant free-riding strategy is close to maximal in groups of type (i), intermediate levels of cooperation are achieved in groups of type (ii), and the lowest cooperation levels prevail in groups with predominantly selfish individuals.

These lab findings are nicely echoed in field evidence. Kosfeld and Rustagi (2015) measured prosocial and antisocial tendencies of village leaders in Ethiopia in a third-party punishment experiment. These leaders were responsible for monitoring and sanctioning of free riders in communities that are strongly reliant on the successful management of forest commons. Rustagi and Kosfeld show that villages with prosocial leaders have significantly better forest outcomes. These results continue to hold after careful consideration of reverse causality issues and omitted variable bias. Similar results of the effects of prosocial preferences on cooperation have been reported in Rustagi, Engel, and Kosfeld (2010) and Carpenter and Seki (2011).

6.2 *Implications of Social Preferences for Labor Relations and Macroeconomics*

6.2.1 *Fairness Concerns, Wage Inequality, and Job Satisfaction*

If people care for equity and reciprocity, it is likely that wage inequalities that violate their fairness standards will have detrimental effects on performance and satisfaction. Many practitioners in human resource management share this viewpoint. Bewley (1999) interviewed several hundred personnel managers about pay-related issues and concluded, for example: "The main function of internal structure is to ensure internal pay equity, which is critical for good morale" (p. 82). However, do *behavioral* data from laboratory experiments and field studies back the views that managers express in surveys and interviews? As we will see, the behavioral data generally provide a strong endorsement for the role of fairness and equity concerns in the assessment of wage inequalities and indicate the conditions under which pay inequalities have detrimental effects on performance.

The notion of inequity aversion implies that employees who work under identical conditions and provide identical effort should be paid identically. If instead they are offered unequal wages, the prediction is that workers who dislike disadvantageous inequality and who receive a lower wage will reduce their effort. Gächter and Thöni (2010) conducted a

laboratory experiment in a repeated one-shot gift exchange setting (i.e., an environment with non-contractible effort) with three players, where one experimental employer faces two workers and makes flat wage offers, w_i and w_j , to each of the two workers. The workers are then informed about w_i and w_j , after which they choose their effort levels, e_i and e_j , which are associated with effort costs of $c(e_i)$ and $c(e_j)$. Gächter and Thöni indeed find that disadvantageous wage discrimination for worker i (i.e., increasing w_j for a given level of w_i) reduces her effort level, while advantageous wage discrimination (i.e., decreasing w_j for a given level of w_i) leaves e_i unaffected.

In subsequent studies (Gächter, Nosenzo, and Sefton 2012, 2013), the authors also showed that social preferences affect workers' effort behavior even in the absence of wage inequality. Their design is similar to that of Gächter and Thöni (2010), but after the workers observed the employer's wage offer they chose their effort levels sequentially. Thus, the effort level of employee 1, who chose first, could affect the effort level of employee 2, who chose second. It turns out that if both workers received generous wage offers, the effort level of employee 2 is strongly positively correlated with the effort level of employee 1—a finding that inequality averse preferences predict.

The lab-based papers just discussed implemented a situation where all parties knew wages, effort levels, and the workers' output at different effort levels. In this setup, the involved parties have clean data that enables them to compare their outcomes. However, what happens if, for example, there are large productivity differences between the workers, but they do not know the exact differences and are only informed that their productivity differs? In this case, social comparison processes are necessarily based on less precise information which—depending on workers' beliefs about their coworker's productivity—may strengthen or weaken the impact of differential wage payments on effort choices. For example, if workers believe that productivity differences are minor, then wage inequality may have negative effects on effort, while they may consider wage differences to be more justified if workers believe that there are large productivity differences. Charness and Kuhn (2007) implemented a three-player gift exchange experiment where one employer faces two workers who do not know the effort–output schedules. They report that workers' effort responded positively to their own wages, which indicates the presence of social preferences, but coworkers' wages did not affect workers' effort choices. A plausible interpretation of this finding is that workers believed that there are large productivity differences that justified differential wage payments.

To what extent are these findings about the negative effort spillovers of coworkers' higher wages generalizable to the field? Cohn et al. (2014) implemented a wage cut in a field experiment that offered a one-time job opportunity to workers who performed the job in teams of two. The firm placed a job advertisement stipulating an hourly wage of about €10 on an online search platform. The task for both workers was identical and consisted of selling promotional cards that permitted entrance to specific nightclubs. The experiment had two phases that were spread over two subsequent weekends with six hours of work per weekend.

In phase one (the first weekend) both workers received the same hourly wage of €12, while in phase two (the second weekend) (i) none of the workers, (ii) only one of the two workers, or (iii) both of the workers received a wage cut of €3 relative to the first weekend. Note that even the workers who received a wage cut on the second weekend earned a higher total income for the overall job ($6 \times 12 + 6 \times 9 = 126$) than that which they initially could have expected with the hourly wage of about €10 initially announced ($12 \times 10 = 120$).

The authors find that if the wage of only one worker in the team is cut, that worker reduces his performance by 34 percent relative to the workers in the no wage-cut group,

while the performance of the worker from the wage-cut group whose wage is not cut remains unchanged.²⁶ Thus, similar to the lab experiments discussed previously, disadvantageous wage inequality is associated with a large negative effect on effort, while advantageous wage inequality leaves effort unchanged. One might therefore expect that a wage cut that affects both workers in a team would lead to a smaller effort reduction, compared to a unilateral wage cut, which is indeed what Cohn et al. observed. If the wage of both workers in a team is reduced, their performance “only” decreases by 15 percent relative to the no wage-cut group, which is statistically significantly different from the 34 percent productivity reduction in case of a unilateral wage cut. These findings are in line with the predictions of a model of inequality aversion.

Breza, Kaur, and Shamdasani (2018, henceforth BKS) implemented a month-long field experiment with Indian manufacturing workers who worked for a daily wage. Their study provides a fascinating and rich collection of facts regarding the relevance of wage inequality for workers’ daily labor supply (i.e., showing up for work), their effort during work, and the impact of wage disparities on the subsequent ability to cooperate in other tasks.

BKS assembled production units consisting of three workers who sit together in a separate physical space during work and lunch breaks. The workers in a production unit thus form a natural reference group. They randomized workers into a pay disparity condition where the daily wage reflected workers’ baseline productivity ranks within the group that was assessed during an initial training period, and into three pay compression conditions: In the “compressed low” condition all three workers received a low wage, in “compressed medium” all received a medium wage, and in “compressed high” all workers received a high wage. This design enables examination of the impact of wage disparity while controlling for absolute wages. And by randomizing production units to different tasks that differ with regard to the observability of coworkers’ output, the authors can examine the impact of output observability on effort during work. In addition, the authors also measured workers’ attendance on the job (they are only paid when they appear on the job) and their knowledge about coworkers’ wages within and across production units. BKS show that workers within a unit are well aware of their coworkers’ wages, while little wage information travels across production units.

BKS find that for workers with a similar baseline productivity and a *given* absolute pay level, a worker’s output declines by 0.33 standard deviations (22 percent) on average when his two coworkers receive roughly 5 percent higher wages. Moreover, if we hold the level of absolute pay constant, there is no evidence that receiving a roughly 5 percent higher wage than one’s peers increases output. In fact, pay inequality in the presence of similar baseline productivities across workers appears to cause a general dissatisfaction with the job situation in the sense that both overpaid and underpaid workers reduce their attendance at the job, compared to the compressed pay condition. This means that the workers in the pay disparity group give up valuable earnings—on average by 9.3 percent—by substantially reducing attendance.

Note that these facts about the effect of over- and underpayment on effort/output nicely coincide with the lab findings of Gächter and Thöni (2010) and the field findings of Cohn

²⁶To interpret this finding, it is useful to understand that while the two workers in a “team” worked during the same shift and in the same environment (e.g., at a well frequented subway station), there was no interdependence in their task and their interactions during a shift were minimal. In particular, they had little information about their coworkers’ effort during the shift.

et al (2014). BKS also show that paying different wages has no negative impact on workers' output in tasks where individuals' productivity differences are easily observable. This suggests that wage inequality has no negative effect when productivity differences justify pay differences—a finding that is consistent with our interpretation of the lab evidence in Charness and Kuhn (2007).

BKS also report a remarkable finding about the detrimental impact of unjustified pay disparities on the subsequent ability of workers to cooperate even when it is in their self-interest to cooperate.²⁷ On the last day of their job, the workers participated in two cooperative games in each of which they could earn money on the basis of group piece rates for performance. The outcome of these games did not affect the firm's payoff. In the first game, the members of a unit had to build towers of raw materials; the higher the tower, the more each worker in the unit earned. In pay disparity units with little or no baseline productivity differences the workers built towers that were 17 percent smaller on average, compared to the compressed pay units. In contrast, when pay differences were justified—based on baseline productivity or task observability—pay disparity units and compressed pay units performed equally well.

Finally, their endline survey reveals that workers from pay disparity units show a significantly lower social cohesion—in terms of their willingness to borrow or lend, or to seek or give advice, or visit one another's homes—compared to workers in the compressed pay group.

More recently, Cullen and Perez-Truglia (2022) conducted a field experiment on the effects of salary comparisons with a sample of 2,060 employees from a large corporation in Southeast Asia. They document substantial misperceptions of managers' and coworkers' wages and identify the causal impact of changes in salary perceptions with the help of an information provision experiment. They find that a higher perceived peer salary has a large negative effect on employee's own effort. A 10 percent increase in employees' perception of peer salaries significantly reduces the number of hours they work by 9.4 percent, the number of emails they send by 4.3 percent, and their sales performance by 7.3 percent. The authors also collected survey evidence that is consistent with the view that social preferences are the mechanism underlying these peer comparison effects, as they also find that higher perceived peer salaries have negative effects on pay and job satisfaction.

These negative effects of peer salaries on job satisfaction are consistent with the findings of Card et al. (2012), who conducted a field experiment with University of California employees by providing them with easy access to information about peer salaries. Employees who had salaries below the peer median subsequently displayed a reduced job satisfaction and an increased intention to switch jobs, while those with salaries above the median showed no changes in job satisfaction and no intention to switch. These findings are consistent with the view that employees have a considerable aversion to disadvantageous inequality.

D'Ambrosio, Clark, and Barazzetta (2018) provide further evidence of the role of fair reference wages on quitting behavior using data from the German Socio-Economic Panel (SOEP). The SOEP contains panel data on which income people consider as fair for their current job. They show that individuals' fair income gap (the difference between what they earn and what they consider fair for their current job) is not only significantly associated with individuals' life and job satisfaction, but it also influences workers' emotional states, such as the frequency of feeling happy or feeling sad and a strong influence on the frequency of experiencing anger.

²⁷ This result is reminiscent of that in Schmitt and Marwell (1972), who find that workers refuse to work in pairs with colleagues who earned more than they did.

Finally, consistent with these findings on subjective assessments of well-being, the fair income gap also predicts the probability of quitting within the next year.

The evidence provided by Dube, Giuliano, and Leonhard (2019) further provides strong support for the view that fairness concerns involving comparisons with (higher) peer wages have a substantial impact on workers' quitting behavior. The authors estimated the own-wage and the peer-wage elasticities of employees' job quitting behavior at a large US retailer with hundreds of stores nationwide. They exploited a regression discontinuity that resulted from the firm's response to the federal minimum wage increases in 1996 and 1997 to identify the causal effect of own and peer wages on quitting behavior. The results show that job separations are extremely sensitive to rising peer wages with peer wage elasticities of 20, 9, and 3 for three, six and nine months after the raise. This result contrasts sharply with the rather low own-wage elasticities they found. Their estimates suggest that, holding the gap between own and peer wage constant, a uniform raise in wages has no impact on quitting behavior. Thus, the overall effect of wages on separations is mostly driven by peer comparisons.

Finally, Dube et al. show that the peer wage effects are asymmetric because they are only driven by comparisons with higher-paid peers, which again suggests that aversion to disadvantageous wage inequality is an important driver of labor market behavior. The overall findings from surveys, lab experiments, and field evidence suggest that aversion to disadvantageous wage inequality that cannot be justified by effort or productivity differences generates strong behavioral effects in terms of a reduced willingness to perform, an increased willingness to quit, a lower job satisfaction, a reduced social cohesion, and lower willingness to cooperate.

6.2.2 *Fairness Concerns and Resistance to Wage Cuts*

There is a considerable literature indicating the importance of fairness concerns for the presence of downward wage rigidity. Surveys conducted by Kaufmann (1984), Blinder and Choi (1990), Agell and Lundborg (1995, 2003), and Bewley (1995, 1998, 2007), as well as the experimental literature related to equity theory (e.g., Lawler and O'Gara 1967), all suggest that workers strongly resist (nominal) wage cuts for fairness reasons even in recessions, and that personnel managers are keenly aware of this resistance. While these surveys do not pin down the concrete social preferences underlying workers' fairness concerns, a plausible interpretation of the evidence suggests that preferences such as negative reciprocity or inequality aversion (with suitable reference points) provide the motivational raw material for these concerns.

Because fairness concerns induce workers to resist wage cuts and personnel managers anticipate this resistance, the survey evidence suggests that firms will be very reluctant to cut wages, which in turn mitigates labor market adjustments to exogenous shocks. There is indeed substantial laboratory and field evidence of downward wage rigidity, and the data often point toward the existence of fairness concerns as the underlying mechanism.

Regarding the field evidence related to whole labor markets, Dickens et al. (2007) document wage rigidity in many countries. Fehr and Goette (2005) show downward nominal wage rigidity for Switzerland, while Grigsby, Hurst, and Yildirmaz (2021) document it for the United States. In addition, Fehr and Goette (2005) show that downward wage rigidity is negatively related to employment, and Kaur (2019) also finds that it is associated with employment distortions. Kaur studies downward rigidity in the context of village labor markets in India and documents strong rigidity. In addition, she reports that Indian village workers consider

nominal wage cuts to be very unfair, suggesting that fairness-related resistance to wage cuts is driving rigidity. Interestingly, workers do not consider *real* wage cuts that arise from avoiding nominal pay raises in response to inflation to be unfair—a finding that is consistent with data reported in Kahneman, Knetsch, and Thaler (1986).²⁸

Firm-level evidence is provided by Greenberg (1990), who reports that workers responded to a temporary wage cut triggered by a negative demand shock with an increase in employee theft during the period for which pay was cut. Krueger and Mas (2004) document that workers at Bridgestone/Firestone's Decatur, Illinois, plant responded to the firm's attempt to cut wages and hire replacement workers with the provision of lower quality tires. Their monthly data show that defective tires were produced primarily during those months in which the firm demanded wage reductions and incumbents worked side by side with replacement workers.

Coviello, Deserranno, and Persico (2022) report evidence on workers' responses to a wage cut in a sales call center in the United States. This company paid its sales representatives on the basis of two performance indicators: commissions based on net sales (gross sales minus refunds due to dissatisfied customers) and conversion rates (percentage of calls resulting in positive gross sales). When the company raised the *required* conversion rates, which was associated with a 13 percent earnings reduction at a given performance, many sales representatives responded by keeping gross sales constant but increasing customer refunds by intentionally selling suboptimal items to the customers. Note that this behavior not only hurt the company but also the workers themselves, indicating that workers were willing to take costly actions to punish the firm for cutting their wages.

Labor relations are often long term. Therefore, if workers respond to wage cuts with reduced effort, sabotage, or higher theft rates, one may interpret this as a rational punishment for employers in a repeated game, that is, the workers' responses may not necessarily result from their social preferences. For this reason, it is useful to study responses to wage cuts in more short-term employment situations in lab and field experiments where there is no prospect for future employment.

The laboratory evidence on downward wage rigidity comes from experimental labor markets (e.g., Fehr, Kirchsteiger, and Riedl 1993; Fehr and Falk 1999; Charness 2004; Brandts and Charness 2004; Brown, Falk, and Fehr 2004) that are designed in such a way that, in the absence of social preferences, employers have an incentive to pay competitive wage levels that are rather low. However, if effort is non-contractible and workers respond to the low competitive wage levels with lower effort (due to social preferences such as reciprocity or inequality aversion), even selfish employers have a pecuniary incentive to pay high, noncompetitive wages. The evidence from these experiments indeed indicates that employers are reluctant to cut wages to competitive levels because they will then receive low effort levels from their workers. The experiments also show that even in the presence of a large excess supply of workers, the experimental employers shy away from cutting wages to low, competitive levels because of anticipated detrimental effects on workers' performance.

Do the negative effort responses triggered by wage cuts in the lab generalize to field experimental settings that credibly rule out repeated game effects? Several studies indicate that the answer is "yes" (Kube, Maréchal, and Puppe 2013; Cohn et al. 2014). We already discussed the

²⁸ Additional behavioral factors, such as money illusion or different responses to acts of commission (such as an explicit wage cut), in contrast to acts of omission (such as an omitted wage increase), are needed to explain the much stronger resistance to nominal wage cuts compared to real wage cuts through omitted wage increases.

negative productivity effects of wage cuts in Cohn et. al. in the previous section. Likewise, the evidence in Kube, Maréchal, and Puppe (2013) indicates a large negative productivity effect of 20 percent from a wage cut relative to a no wage-cut treatment.

Another interesting study documenting downward wage rigidity and the associated employment effects is Breza, Kaur, and Shamdasani (2021). These authors implemented hiring shocks in local Indian labor markets by giving jobs in *external* jobsites to an average of 24 percent of the labor force of casual male workers for two to four weeks—a shock that substantially reduced how many workers remained in the local economy.

Their approach exploits the strong seasonality in labor demand in these local labor markets. The hiring shock led to immediate and strong raises in wages and a fall in employment in the local markets during the peak season, when demand for casual labor is generally high. However, the hiring shock had basically no impact on employment and wages in the local market in the lean season, when the demand for casual labor is generally low. This is a remarkable finding, since there were apparently enough unemployed workers to fill the gap generated by removing 24 percent of the available labor force, indicating severe rationing of labor supply—a finding that could not have happened in a competitive labor market with flexible wages. If the local labor market during the lean season had been cleared before the hiring shock, then the shock also should have led to large wage increases and a reduction in employment. However, there were apparently enough unemployed workers before the hiring shock who were willing to work at the going wage but who could not find employment at that wage. This made it possible for employers to find enough workers without having to raise wages despite the considerable reduction in the local labor force. Breza, Kaur, and Shamdasani also provide evidence indicating that moral hazard or nutrition efficiency wage models cannot explain their data, while a model that relies on workers' resistance to wage cuts can.

Quach (2020) provides evidence for downward wage rigidity by exploiting the following natural experiment from the United States. In May 2016, the federal Department of Labor announced that starting December 1, 2016, salaried workers earning less than \$913 per week would be entitled to overtime compensation if they work more than 40 hours in a week. In response to this announcement, many employers promised raises to their employees in anticipation of the new rule. However, one week before the rule became effective, a federal court ordered an injunction on the new policy, implying that the employers would not face any legal obstacles if they wanted to refrain from the promised pay raises.

Quach (2020) shows that employers nevertheless increased wages. For the median worker, for example, wages rose by 5.8 percent, suggesting the employers shied away from cutting nominal wages to preannouncement levels. The pay raises took the form of bunching many employees at \$913 per week and reclassifying some workers from salaried pay to hourly pay. Quach also shows that workers who received pay raises through bunching experienced the same wage growth as workers slightly above the bunching threshold, suggesting that firms did not lower the future wage growth of workers whose wages exhibited rigidity. Moreover, the paper shows that even a year after the proposed overtime policy was nullified, the employers continued to bunch the salaries of new hires at the \$913 threshold, indicating that nominal wage rigidity also affected the new hires.

The evidence from Quach (2020) suggests that even the mere promise of a pay raise based on a *temporary* legal requirement makes it hard for employers to subsequently lower wages. This finding is also consistent with the laboratory evidence documented in Falk, Fehr, and

Zehnder (2006), where the temporary implementation of a legal minimum wage led to lasting effects on wages that prevailed even long after the legal minimum wage was removed.

6.2.3 *Screening and Selection Based on Social Preferences*

If social preferences are a relatively stable individual attribute, employers may want to attract workers with particular social preferences and avoid workers with others. Workers with altruistic preferences, for instance, may be valuable for employers because they generate positive spillover effects on other workers in interdependent production processes. Conversely, employers may shy away from workers with envious or spiteful social preferences because they may have detrimental effects on cooperation among employees and between the envious employee and the employer. Likewise, employers might avoid workers who are negatively reciprocal because they may have a strong tendency to engage in counterproductive activities when they are aggrieved.²⁹

There is a relatively large literature suggesting that people with more prosocial inclinations tend to self-select themselves to a higher degree into the public sector in countries with a high degree of trust into the public sector (e.g., Dur and Zoutenbier 2014). Most of these studies are based on self-reported data about motivation or self-reported prosocial actions. However, evidence based on revealed preference data also exists. Buurman et al. (2012) show that early career public-sector workers are more likely to donate to a charity compared to observationally equivalent private-sector workers. Gregg et al. (2011) study British Household Panel Data and show that workers who are more prosocial—in terms of providing unpaid overtime work—are more likely to sort into the nonprofit sector. They also find that this effect is strongest for industries with “caring characteristics,” such as health, education, and social care.

Prosocial individuals may not only prefer working in companies and sectors with caring or helping characteristics, but they may also shy away from sectors or companies involved in immoral business practices, such as the intentional sale of toxic financial assets, the marketing of tobacco products to underage smokers, or the aggressive marketing of opioids by the pharmaceutical industry. Schneider, Brun, and Weber (2024) used administrative, laboratory, and survey data to study the hypothesis that the least prosocial (i.e., most immoral) people are most likely to work in jobs perceived to involve (or actually involving) immoral activities. Moreover, if working for a company/industry that is perceived to be involved in immoral activities is emotionally aversive, standard economic theory would predict compensating wage differentials. In other words, labor market competition would induce companies/industries perceived to be more immoral to pay, *ceteris paribus*, higher wages.

To examine the compensating wage differentials hypothesis, they collected survey data from the Swiss population on the perceived morality/immorality of different industries in Switzerland. Then they regressed the gross hourly wages across the industries on the industries’ perceived immorality, controlling for observable industry and workers’ characteristics. The results indicate a strong positive correlation between the perceived immorality and the gross hourly wages, with industries such as tobacco and weapons manufacturing paying the highest wages and construction and sports facilities being among the lowest paying industries.

²⁹ Selection and sorting only make sense if individuals’ social preferences or their assignment to a particular social preference type exhibits a reasonable degree of stability over time. In Supplemental Appendix 5, we discuss evidence suggesting that this is the case.

Because the correlational evidence from administrative data is, of course, not yet fully convincing, they exogenously varied the characteristics of competitive experimental labor markets. In the immoral work treatment, the subjects were competing for jobs that required them to give wrong advice to another individual that reduced that individual's earnings *and* charitable donations to UNICEF. In the moral treatment, the job involved giving advice that increased another individual's earnings and donations to UNICEF.

The striking result of this experiment is that the reservation wages, and thus the competitive equilibrium wages, for the immoral job are much higher than for the moral job. Moreover, subjects with weaker prosocial preferences have a much higher frequency of employment in the market involving the immoral task. Thus, the lab experiments provide causal evidence for compensating wage differentials for immoral jobs and for selective sorting of more immoral individuals into these jobs. Finally, the authors also show, with the help of survey evidence, that subjects who are less prosocial are more willing to work for industries perceived to be more immoral.

Dohmen et al. (2009) provide further evidence on the sorting/selection hypothesis of social preferences. Their results suggest that workers' attitudes toward positive and negative reciprocity can have quite far-reaching effects on their workplace behavior and their earnings. They exploit an interesting survey measure of reciprocity (Perugini et al. 2003) that was included in the German Socio-Economic Panel (SOEP) in 2005, which enabled them to estimate the association between workers' willingness to voluntarily provide effort (in the form of overtime work) and their positive and negative reciprocity.

Controlling for a large number of individual characteristics, Dohmen et al. show that positive reciprocity measured in 2005 is significantly associated with workers' actual overtime work in the years 2005 as well as in the years 2006 and 2007. Moreover, the coefficient on positive reciprocity is almost twice as large for workers who perceived their current wage as fair, while if workers perceive their current wage as unfair, the association between overtime work and positive reciprocity is zero. These results are consistent with the view—derived from theories of inequity aversion and reciprocity—that wages perceived as fair induce reciprocal workers to increase their work effort.

Dohmen et al. also find that negatively reciprocal workers are less willing to perform overtime work. In addition, they show that positively reciprocal workers are less likely to be absent from the workplace, while negatively reciprocal workers tend to be absent more often. Likewise, positively reciprocal workers “consume” fewer days for paid sick leave, while negatively reciprocal workers are on paid sick leave for more days.

Based on these results, one would expect that positively reciprocal workers are more valuable employees, that is, that the labor market will reward them with higher wages, while negatively reciprocal workers are less valuable. Dohmen et al. (2009) estimate Mincer-type wage equations and indeed find that positively reciprocal workers earn higher monthly and annual labor incomes. They do not find a negative impact of negative reciprocity on wages, but instead they show that negatively reciprocal workers have a higher probability of being unemployed, while positive reciprocity reduces the probability of being unemployed.

Barr and Serneels (2009) conducted trust experiments with several hundred employees from 20 manufacturing companies in Ghana; 164 of them were in the role of the second mover, which provides a (noisy) measure of the workers' willingness to display reciprocal behavior. Note that the first mover in this experiment reaps a positive rate of return if she gets back more than what she transferred. Barr and Serneels categorize a worker as highly

reciprocal if his back-transfer yields a rate of return of more than 50 percent for the first mover.³⁰ The authors show that the output per worker across companies is strongly positively correlated with the share of high reciprocators among employees. This correlation persists when controlling for capital inputs and sector fixed effects. Moreover, a Mincer-type earnings regression that includes a dummy for highly reciprocal workers indicates that these workers earn a wage premium.

Although the papers by Dohmen et al (2009) and Barr and Serneels (2009) do not establish a causal relationship between workers' willingness to reciprocate and their workplace behaviors and earnings, they nevertheless constitute suggestive correlations that deserve further scrutiny. Their findings are consistent with what one would theoretically expect based on knowledge about the behavioral properties of the involved social preferences, and they are also consistent with the literature on the impact of early childhood characteristics on later life outcomes. Vergunst et al. (2019) show, for example, that teachers' ratings of kindergarten boys' prosociality are positively associated with the boys' earnings in adulthood, after controlling for a large set of covariates.

The findings in Dohmen et al. and Barr and Serneels are also in line with causal laboratory evidence of Bartling, Fehr, and Schmidt (2012) on the role of employers' screening in experimental labor markets. In these experiments, employers in some treatments can condition their job offers—in terms of wages, rent sharing, and employees' opportunities for effort discretion—on information about employees' past performance levels in other firms. The experimental employers make ample use of this information and offer completely different compensation packages to the workers depending on their past performance. Workers with high past effort levels—generally based on their willingness to reciprocate to generous job offers—receive generous current job packages with high wages, a high share of the overall surplus, and broad opportunities for effort discretion. In contrast, workers with low past performance received mediocre job packages with low wages, no rent sharing, and tightly controlled effort opportunities. These findings suggest that, under the realistic assumption that employers can acquire information about their employees' effort attitudes, positively reciprocally motivated workers are rewarded with better job packages.

6.3 *The Role of Social Preferences for Incentives, Contracts, and Institutions*

6.3.1 *The Effects on Contract Enforcement and Financial Incentives*

Over the past 20 years, several authors have illustrated the advantages and disadvantages of different incentive schemes in the light of fairness concerns. For example, it has been shown theoretically (e.g., Sliwka 2007) and experimentally (e.g., Fehr and Gächter 2002; Fehr and List 2004; Falk and Kosfeld 2006) that explicit incentive contracts may undermine voluntary cooperation based on social preferences. As a consequence, explicit incentive contracts may be less efficient than implicit alternatives based on trust, informal bonuses, or informal sanctions.

Fehr, Gächter and Kirchsteiger (1997) and Fehr and Gächter (1998), for example, tested the impact of trust and reciprocity on contract enforcement in a standard one-shot gift exchange

³⁰The parameters of the trust experiment are such that at a 100 percent return for the first mover, the payoffs between the two parties are equal, while the second mover reaps all the surplus generated from the first mover's transfer at a zero rate of return.

experiment where principals commit to pay a wage and state a desired effort level in stage 1 and workers respond to the offer with an effort choice in stage 2. In an additional treatment they added a third stage in which the experimental firms can pay to reward or punish the worker for her effort choice. There was a positive wage–effort relation in both treatments, but the average effort level was much higher in the three-stage treatment than in the two-stage treatment. Thus, workers apparently anticipated that firms reward high-effort choices and punish low ones, indicating that opportunities to informally reward or sanction workers, which in reality almost always exist, can have powerful incentive effects even in one-shot interactions.

Fehr, Klein, and Schmidt (2007) provided evidence suggesting that social preferences may induce principals to prefer informal bonus contracts over more formal contracts with explicit incentives. They consider the following three types of contracts: a trust contract in a two-stage gift exchange environment like the one described above, an incentive contract that introduced an explicit incentive into the trust contract, and a bonus contract that introduced the option of informally rewarding agents in a third stage. The incentive contract is based on a verification technology that enables the principal to fine the agent in case of verified shirking. The verification technology is imperfect and the fine is limited, implying that the highest effort level that can be implemented is positive but falls short of the efficient effort level. The informal bonus contract contains no explicit incentives, but gives the principal the opportunity to reward agents *ex post*, that is, after effort is observed. The bonus contract does not rely on effort verification and enforcement by third parties. Instead, the principal promises a nonbinding, voluntary bonus payment if the agent's effort is satisfactory. This bonus contract is an implicit contract because third parties do not enforce the principal's promise.

If all actors were completely selfish, the incentive contract is the only viable contract, and the trust and bonus contracts would be equally bad. However, the incentive contract in fact dominates the trust contract, but the bonus contract turns out to be much more efficient than the incentive contract. How is it possible that social preferences are not strong enough to render the trust contract more efficient than the incentive contract, but strong enough to make the bonus contract the most efficient one? Fehr, Klein, and Schmidt (2007) show that inequality aversion preferences can explain this puzzle.

6.3.2 *Social Preferences as a Behavioral Foundation for Employment Contracts*

The existence of simple employment contracts that pay a state-independent fixed wage and give employers the right to tell the employee what to do (i.e., to exert authority) is a long-standing puzzle in economics. Why should the trading parties ever agree to such a seemingly inefficient contractual arrangement that may prevent trade (i.e., employment) in certain states of the world? Why do they not continuously renegotiate the contract terms, allowing them to respond to changing conditions and achieve *ex post* efficient outcomes, as Alchian and Demsetz (1972) suggest? And if continuous and efficient *ex post* renegotiation is always possible, isn't the characterization of the employment contract as an authority relation thoroughly misguided? A negotiation, after all, means that task assignments are subject to *both* parties' agreement.

Hart and Moore (2008) tackle these and related questions by dropping the assumption that "*ex post* trade is perfectly contractible" and that "renegotiation always leads to *ex post*

efficiency” (p. 3).³¹ In the absence of perfect ex post contractibility, we are in the world of incomplete contracts with gift exchanges and informal relationships where social preferences typically play a key role. Moreover, social preferences and their interactions with contractual arrangements deeply affect ex post inefficiencies (Fehr, Gächter, and Kirchsteiger 1997; Fehr, Klein, and Schmidt 2007), and thus also determine which contracts are most efficient. This raises the question of whether social preferences could also render employment contracts with rigid wages more efficient, compared to contracts that allow for the flexible adjustment of wages to the prevailing state of the world.

Fehr, Hart, and Zehnder (2011) indeed show experimentally that rigid contracts can be superior to flexible contracts. In the experiment, there is ex ante uncertainty whether a good state of the world (e.g., high output prices) or a bad state of the world prevails. In addition, the parties’ values and costs in the different states of the world are not verifiable, so that state-contingent contracts cannot be written. The advantage of a flexible contract that fixes only a wage *range* but not the wage *level* is that it allows adjusting the wage, w , such that trade between an employer and an employee is also possible when output prices are low. In contrast, a contract with wages that are rigidly fixed ex ante (i.e., before the state of the world is known) may prevent trade in this situation. The flexible contract may, however, also be disadvantageous because it provides scope for diverging expectations regarding the wage that will be paid ex post. In other words, while the rigid contract pins down wage expectations ex ante and thus avoids ex post disappointments, workers under flexible wages may feel entitled to higher ex post wages in a good state of the world, which provides scope for ex post disappointments. In the presence of non-contractible effort levels and fairness concerns (social preferences), workers may thus shirk more under flexible contracts than under rigid contracts. Moreover, the lower effort levels under flexible contracts may even render that contract less profitable than the rigid one.

The experimental results confirm the conjectures above. The drawback of the rigid contract is that it prevents trade in the bad state of the world, but this is often overcompensated by the fact that rigid contracts elicit considerably higher effort levels in the good state of the world. To prevent disappointments and low effort levels in the good state of the world, employers pay much higher ex post wages under flexible contracts but still observe a non-negligible amount of shirking. In contrast, the wage is competitively fixed at very low levels by market competition under rigid contracts, and low effort levels rarely occur despite these low wages.³² Overall, this renders fixed wage contracts more profitable than flexible contracts. Note that this result could not occur with selfish actors because the flexible contract would always dominate the rigid one.

6.3.3 *Social Preferences, Contractual Incompleteness, and Property Rights*

Many investments are relationship specific, meaning that they are valuable only within a particular relationship. In the presence of incomplete contracts, these investments bear the risk of being exploited ex post—the so-called hold-up problem. Rational parties anticipate being held up ex post, and thus underinvest in relation-specific assets. The property rights

³¹ Hart and Moore (2008, p. 3) ask the following fundamental question: If the relevant parties can always sit down together ex post (i.e., after the state of the world is revealed) and bargain to an efficient outcome, why should “authority, hierarchy, delegation, or indeed anything apart from asset ownership matter”?

³² Under flexible contracts, competition only determines the lower bound on wages.

literature shows that the appropriate allocation of asset ownership can mitigate the underinvestment incentive. Incomplete contracting and the associated hold-up problem have thus provided an important economic rationale for the allocation of asset ownership to those parties who are the most vulnerable to exploitation (Grossman and Hart 1986; Hart and Moore 1990).

The key assumption behind the property rights approach is that contracts are incomplete, which is justified by assuming that payoff-relevant information is observable to the involved parties but not verifiable by a third-party enforcer. There are many other applications of the “observable but not verifiable assumption” in economics and the assumption has therefore become one of the most important cornerstones of modern institutional economics.³³ However, all these applications of the incomplete contracting approach are subject to a fundamental criticism that Maskin and Tirole (1999) have raised. They show that if parties commonly observe payoff-relevant information, one can construct an extensive form mechanism that leads to truthful revelation of the relevant information in the *unique* subgame perfect equilibrium of the game the mechanism implies.

Thus, one could in principle design contracts that embody this extensive form mechanism, and if the mechanism works as theory predicts, all commonly observable information could be turned into truthfully reported verifiable information. This means that the *second-best* institutional arrangements derived under incomplete contracting would become superfluous because a superior contractual arrangement exists. The question, however, is whether the extensive form mechanism mentioned above, which is based on the work of Moore and Repullo (1988), indeed works as predicted.

Fehr, Powell, and Wilkening (2021) examine this question experimentally and show that negative reciprocity thoroughly undermines the functioning of Maskin–Tirole-type mechanisms. Most parties are unwilling to enter a contract that incorporates the mechanism, and if they enter them, these contracts typically perform worse than contracts without the mechanism. Intuitively, a key reason for the failure of the mechanisms is that they are based on large fines for the trading parties if they “misbehave,” but the threat and execution of large fines is also likely to induce extreme hostility (i.e., negative reciprocity) between the parties. Adding the mechanism to a usual hold-up problem is like handing out guns at a fist fight. The guns are unlikely to make the fight more peaceful.

Thus, social preferences in the form of negative reciprocity undermine the criticism of the theoretical foundations of incomplete contracting models. Ironically, to the extent to which social preferences are a force that contributes to contractual incompleteness, they help sustain their own behavioral importance. Why? Because incomplete contracts provide the terrain—gift exchanges, informal sanctions and rewards, informal agreements—under which social preferences can play an important role.

Overall, social preferences may contribute to the prevalence of incomplete contracting in two ways. First, they may render institutions like the mechanisms just discussed, that render contracts more complete, dysfunctional. Second, they may mitigate the contracting problems that arise under incomplete contracts. One example of this is the relatively high efficiency of incomplete bonus contracts in Fehr, Klein, and Schmidt (2007), which we have already discussed. Another example is provided by the large experimental literature on behavior under

³³The assumption has, for example, been used to understand property rights and firm boundaries, the optimal scope of governments, problems of privatization, the control of insiders by outsiders through voting rights, financial contracts, and patterns of international trade and technology adoption.

the hold-up problem (see Yang 2021 for a review). This literature shows that the underinvestment problem is typically considerably less severe than the self-interest model predicts (see, e.g., Gantner, Güth, and Königstein 2001; Ellingsen and Johannesson 2004a, b; Dufwenberg, Smith, and Van Essen 2013). A key reason for this is that fairness concerns induce the parties to take their ex ante investments in the ex post bargaining process partially into account. This means that the investing parties experience less exploitation than predicted under self-interest, which weakens underinvestment. Negative reciprocity and disadvantageous inequality aversion are also potentially important forces in this context because they provide a preference-based commitment to credibly reject very unfair offers. Parties who can hold up their counterparts thus often face the threat of complete disagreement (like in the simple ultimatum experiment), and therefore they shy away from fully exploiting their ex post bargaining power.

6.4 *The Role of Social Preferences in Politics*

One of the most treasured public goods is the existence of democratic rights, such as the freedom of speech, the freedom to found political organizations, and the freedom of all citizens to participate in competitive elections. However, these democratic rights did not just fall from heaven; instead, democratic mass movements fought—sometimes for decades—for the institutionalization of these rights. Who are the people who fight for these rights and what are their social preferences? A recent paper by Cantoni et al. (2022) shows that individuals' social preferences and their risk tolerance are the strongest predictors of protest participation in Hong Kong's antiauthoritarian movement. This result holds for university students and the broader population and for modest as well as massive protests. Findings like these suggest that we owe the existence of democratic rights to the prosocial individuals who were willing to take the risks of protesting and fighting against dictatorships.

Yet, the effect of social preferences in politics goes beyond protest participation. Inequality aversion, quasi-maximin preferences, and altruism imply that individuals care about payoff distributions, which should affect their willingness to vote for redistributive policy proposals. Suppose, for example, an economic environment like that in the famous model of Meltzer and Richard (1981), but assume that people are inequality averse. In this environment, individuals face the decision to vote on a proportional tax rate, τ , to be levied on all individuals in the population and redistributed equally as a lump sum. An individual with gross income y_i who receives a lump sum transfer, T , will thus have a consumption level of $c_i = (1 - \tau)y_i + T$. Assuming that there are quadratic resource costs of taxation of $\tau^2/2$ per tax dollar, the government budget is balanced if $T = (\tau - \tau^2/2)\bar{y}$, where $\bar{y}$ represents the average gross income in the population. If one assumes that individuals have Fehr–Schmidt preferences, the preferred tax rate τ^* (for an interior solution) is given by

$$(13) \quad \tau_i^* = 1 - \frac{1}{\bar{y}} \left(y_i - \bar{\alpha}_i \frac{1}{n-1} \sum_{j \neq i} \max\{y_j - y_i, 0\} - \bar{\beta}_i \frac{1}{n-1} \sum_{j \neq i} \max\{y_i - y_j, 0\} \right),$$

where $\bar{\alpha}_i$ captures the aversion to disadvantageous inequality (“envy”) and $\bar{\beta}_i$ the aversion to advantageous inequality (“empathy”). The above solution for τ_i^* suggests that the preferred tax rates for a selfish and an inequality averse individual may look like those in Figure 4 below.

More precisely, the equation (13) yields the following predictions and implications for empirical research: (i) Low-income individuals obviously have a selfish reason for choosing

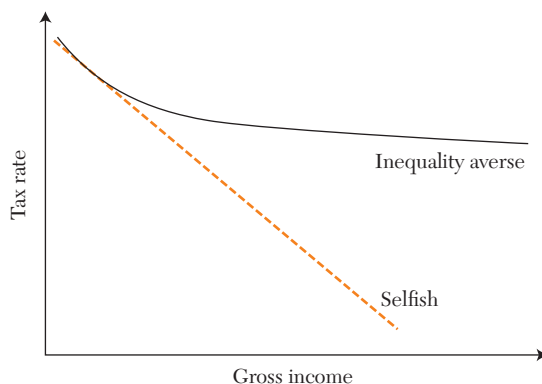


Figure 4. Preferred Tax Rate as a Function of Gross Income and Social Preferences

redistributive taxation, that is, even in case of $\bar{\alpha}_i = \bar{\beta}_i = 0$ they favor a high tax rate. (ii) Inequality averse individuals ($\bar{\alpha}_i > 0$, $\bar{\beta}_i > 0$) have a higher preferred tax rate than selfish individuals ($\bar{\alpha}_i = \bar{\beta}_i = 0$), and those with a higher (positive) $\bar{\alpha}_i$ will demand more redistribution. Likewise, those with a higher (positive) $\bar{\beta}_i$ will demand more redistribution. (iii) A higher gross income y_i will generally lower the demand for redistribution, but this effect will be mitigated for inequality averse individuals. In fact, the term that multiplies $(1/\bar{y})$ in equation (13) may become close to zero for very inequality averse individuals, implying that their demand for redistribution does not decline much with gross income.

Several lab studies on political redistribution—such as those of Tyran and Sausgruber (2006) or Durante, Putterman, and van der Weele (2014)—indicate that social preferences play a role in voting decisions. Durante, Putterman, and van der Weele, for example, assembled groups of 21 subjects whose pretax incomes were calibrated to proportionally reproduce the actual US pretax income distribution. Each subject made a choice regarding the preferred proportional tax rate $t \in \{0.1, 0.2, \dots, 0.9, 1\}$ that generated tax revenue that was equally redistributed as a lump sum to all 21 members of the group. The final group outcome was not determined by voting but by the decision of one randomly chosen subject from the group. This has the advantage that every subject's decision had the same probability of being decisive, that is, incentive compatibility also held for subjects with extreme preferences. The study shows that most subjects are willing to pay to reduce income inequality among others, but they also take the direct costs of taxation and deadweight losses into account when voting on tax rates. Durante et al. also estimate the parameters of a Charness and Rabin model and find that the weight given to increasing the income of the worst-off player in the group is about three times higher than the weight given to aggregate earnings.

To what extent are individuals' social preferences predictive of their political behaviors outside the laboratory, that is, their preferences for left- versus right-wing parties and their demand for redistribution in the broader society? Kerschbamer and Müller (2020) measured the social preferences of a large representative sample of the German population (the German Internet Panel, GIP, see Table 1) that also contained various questions indicating individuals' views on redistribution spread over several survey waves. These are questions like “Should the government mitigate income differences?” or “Should people, who work more

and consequently earn more, pay more or less taxes than they currently do?” Kerschbamer and Müller show that, compared to selfish subjects, inequality averse and altruistic subjects have (i) a higher propensity to vote for left-wing parties, (ii) self-report that they are more left leaning, and (iii) are more in favor of redistribution as measured by the first principal component of the bundle of redistribution questions in the GIP.

Epper et al. (2024) measured the social preferences of a large representative sample of roughly 9,000 Danish individuals together with a survey measure of support for public redistribution. In addition, their dataset contains third-party verified information on individuals' incomes, wealth, education, their grade point average at age 16, further demographic controls, beliefs about the causes of income inequality, as well as incentivized measures of individuals' risk and time preferences. Controlling for this large number of covariates, the authors show that experimental measures of (i) aversion to disadvantageous inequality and (ii) aversion to advantageous inequality are independent predictors of support for public redistribution—a result that fully supports the predictions implied by equation (13) above. Moreover, their results indicate that, to avoid biased estimates, it is important to simultaneously include both measures of inequality aversion in the analysis because advantageous and disadvantageous inequality aversion are positively correlated.

Yet there could be several other potential reasons why people might be for or against redistribution that are not captured in Kerschbamer and Müller (2020) and Epper et al. (2024). These are variables such as individuals' expected future income, their history of misfortunes (i.e., unemployment or negative health shocks), their (potentially false) beliefs about the prevailing inequality, their beliefs about their relative incomes, or their beliefs about the role of luck and effort for economic success in life. All of these reasons have been intensely discussed and examined in the political economy literature on redistribution, which raises the question of whether distributional preferences are also predictive of people's demand for redistribution if one controls for these motives. In addition, there is the question of the extent to which answers to non-incentivized survey questions, such as whether the government should mitigate income differences, validly capture the demand for redistribution.

Fehr@Epper@Senn (2023b) tackle these problems by measuring social preferences in a broad sample of the Swiss population for whom they also elicit measures of the motives for redistribution mentioned in the previous paragraph that allows them to control for these motives. They asked people for the intensity of their support of several strongly redistributive referenda that were put to vote under the rules of Swiss direct democracy during the last 10–12 years. These survey results were then validated with the actual voting results by comparing the geographic and sociodemographic distribution of votes with the distribution of survey answers. In addition, they validated the survey results with people's actual donations to organizations that support or oppose redistributive proposals.

Fehr@Epper@Senn (2023b) document that differences in the support for redistribution across the different social preference groups (selfish, inequality averse, and altruistic) is very small at low incomes but rather large at higher incomes. Both altruistic and inequality averse individuals with above-median incomes display much more support for redistribution than selfish individuals. For example, the support for redistribution among inequality averse individuals with incomes above the median is 0.57 standard deviations higher than the support of selfish individuals. These results follow from the fact that the support for redistribution declines sharply with increasing income for selfish individuals but, as depicted in Figure 4, social preferences strongly mitigate this decline in support for redistribution.

The above results do not, however, yet exhaust the role of social preferences in the demand for redistribution. The reason for this is that both the Kerschbamer and Müller paper and the Fehr@Epper@Senn paper used a distributional measure of social preferences that does not account for people's concern for meritocracy. However, as discussed in Section 3.2, people with meritocratic concerns care about other people's incomes and have, therefore, social preferences, while selfish individuals show no concern for meritocracy (see equations (9) and (10) in Section 3.2).

Starting with Fong (2001), there is a sizeable literature that shows that people's beliefs about the role of effort and luck in economic success is a key factor in their demand for redistribution (Alesina and La Ferrara 2005; Alesina and Giuliano 2011). This literature also comprises studies that examine how beliefs in intergenerational mobility and equality of opportunity affect the demand for redistribution (e.g., Alesina, Stantcheva, and Teso 2018). Individuals who believe that equality of opportunity already prevails, that is, that effort (and not luck) is a primary driver for success in life, often attribute low income to a lack of effort. In other words people with low income are considered responsible for their situation and do not deserve help through redistributive legislation. The combination of beliefs about the important role of effort with meritocratic concerns thus reduces the demand for redistribution. Moreover, because selfish individuals have no meritocratic concerns, beliefs about the role of effort should affect other-regarding individuals' demand for redistribution while selfish individuals should remain unresponsive to these beliefs. This is exactly what Fehr@Epper@Senn (2023b) found.

Thus, taking the meritocratic dimension into account leads to a more nuanced view about the role of social preferences in redistributive politics. The widespread existence of meritocratic other-regarding preferences has been documented not only in the laboratory but also in several survey experiments conducted with general population samples (Almås, Cappelen, and Tungodden 2020; Cappelen et al. 2022), and can, in particular, also explain the popularity of workfare programs (Fong, Bowles, and Gintis 2005; Drenik and Perez-Truglia 2018). Survey experiments have also documented how social preferences and demand for redistribution in general, and meritocratic preferences in particular, are sensitive to the (mis)perceptions a person holds about society, and her own place (e.g., rank) in it (Cruces, Perez-Truglia, and Tetaz 2013; Karadja, Mollerstrom, and Seim 2017; Fehr, Mollerstrom, and Perez-Truglia 2022).

7. Outlook

Over the previous two to three decades a lot has been learned about the properties, prevalence, and consequences of social preferences, but there is also still a lot that needs to be learned. In the following, we outline several open questions that offer exciting research opportunities.

Perhaps, the most fundamental question is related to the determinants of social preferences. There is already initial evidence suggesting that they are formed in childhood through different role models or different early childhood education practices (Van Lange et al. 1997; Cappelen et al. 2020; Kosse et al. 2020) but the set of societal determinants is probably much larger. For example, is it possible for companies and other organizations to shape the social preferences of their employees by structuring rewards, incentives, and the overall company culture in different ways? What is the effect of detrimental health and income shocks on social preference? How does the break-up of marriages or other events that disrupt or improve the relation between family members or members of a community affect social preferences? A

recent paper (Cassar et al. 2022) suggests, for example, that allomaternal care increases prosocial preferences in a community; and Rao (2019) shows that having poor classmates makes rich students more prosocial, generous, and egalitarian and less likely to discriminate against poor students.

We also know very little about how a society's governance and economic institutions shape individuals' social preferences, although Agneman and Chevrot-Bianco (2023), Enke (2023), and Rustagi (2024) have recently made important advances. While the first two papers provide evidence that market participation/integration of communities contributes to prosocial preferences, Rustagi shows how a history of self-government and democratic interactions tends to favor preferences for cooperation. Rustagi exploits a natural experiment in Switzerland, where during the Middle Ages, the absence of an heir resulted in the extinction of a prominent noble dynasty, which enabled some Swiss municipalities to become self-governing whereas others remained under feudalism for another 600 years. Rustagi shows that individuals from self-governing communities display stronger preferences for conditional cooperation (measured in a behavioral experiment) as well as higher voter turnout and higher charitable donations.³⁴ These findings are also consistent with those of Guiso, Sapienza, and Zingales (2016), who show that northern Italian cities that experienced a period of independence in the Middle Ages have significantly higher prosocial behaviors in terms of organ/blood donations, the frequency of cheating in national exams taken by children in each Italian town, and the number of nonprofit organizations.

When discussing the potential determinants of social preferences, the relationship between intrinsic social preferences and social norms may also become important. We define a social norm as a commonly known standard of behavior that is based on a widely shared view of how individual group members *ought* to behave in a given situation (Fehr and Schurtenberger 2018). Thus, in contrast to preferences, which are a property of individuals, social norms are a property of whole groups of people. They constitute an external normative constraint on individuals' behavior that arises from the fact that the normative standard is widely shared and deviations from the standard are met with disapproval, ridicule, and other forms of sanctioning. However, over time external normative constraints may be internalized, which turns them into preferences, but very little is known conceptually and empirically about these internalization processes, the factors that shape them, and ways to model them (although see Enke 2019; Schulz et al. 2019; Ellingsen and Mohlin 2022). In addition, there is very little empirical research that simultaneously elicits and measures social norm-driven and social preference-driven behaviors (for an exception see Carpenter and Robbett 2022).

Another important unresolved question concerns the determinants of individuals' reference points for their fairness and equity judgements. In the absence of reliable empirical knowledge, models like those of Fehr and Schmidt (1999) have pragmatically assumed that (at least in experiments) equality between the involved parties is a good first-order approximation of an individual's actual reference point. As the section on the role of merit, luck, and risk in social preferences has made clear, however, there are many situations in which equality may be the wrong reference point for many individuals. It is therefore important to develop methods that enable the reliable empirical identification of individuals' reference agents and reference

³⁴ Because Switzerland tracks every family's place of origin in registration data, Rustagi can identify the "cultural origin" of individuals and document the persistence of cultural transmission at the individual level in a context of historically low migration rates.

outcomes. An interesting step in this direction has recently been undertaken by Hvidberg, Kreiner, and Stantcheva (2023) and Xu et al. (2023).³⁵

Finally, it would be desirable to study the deeper implications of heterogeneous social preferences for normative (public) economics. If individuals display altruistic or inequality averse distributional preferences, it does not make much sense to compute optimal policies on the basis of social welfare functions that assume that every individual only cares for his or her own consumption. Isn't economics, after all, built on a deep commitment to respect individuals' preferences? Likewise, if people care also for equality of opportunity, it appears of paramount importance to incorporate that notion into modern welfare economics rather than computing optimal policies on the basis of a standard utilitarian welfare function that assumes that individuals only care for their own consumption. In a recent American Economic Association Distinguished Lecture, Emmanuel Saez (2021) echoed this view by stressing the importance of concerns about inequality, poverty, and relative position for positive and normative public economics.

Some steps in this direction have been undertaken by Aronsson and Johansson-Stenman (2020a), who have studied optimal income taxation in the presence of externalities and inequality averse individuals, and derived optimal second-best taxation conditions when individuals have social preferences (Aronsson and Johansson-Stenman 2020b). And more recently, Eden and Piacquadio (2023) discussed the normative content of other-regarding preferences. The generalized social marginal welfare weights approach to optimal taxation (Saez and Stantcheva 2016) also opens up ample opportunities to link insights from social preference research with (normative) public economics for the purpose of deriving optimal tax and transfer policies. However, to establish this link, it is important to move beyond bilateral experimental settings and offer subjects the opportunity to redistribute resources across the whole income distribution. Capozza and Srinivasan (2024) and Fisman, Kuziemko, Vannutelli (2021) recently made interesting steps in this direction. Capozza and Srinivasan estimate the welfare weights of a representative sample of the US population by eliciting individuals' willingness to redistribute money among US citizens from different income classes. They find that the general population weights are more progressive than the weights implied by current tax and transfer policies, suggesting that the general population desires additional redistribution. Fisman, Kuziemko, and Vannutelli (2021) show that when people are given the option of changing the income distribution, they exclusively care for the very poor (positively), the very rich (negatively), and their local "income neighbors" directly above them (negatively). Overall, the role of social preference research for normative and positive public economics may be substantial and, perhaps, change what economists recommend to policymakers.

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³⁵There exists also an older literature in labor economics that discussed reference points such as one's own past wages, peer wages in the company, the company's ability to pay, workers' perceived contributions, etc. as potential reference points (e.g., Levine 1993). It is, however, probably fair to say that no firm conclusions have been reached by this literature.

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